Reviewer-Revised Research Report: Transformation Vectors and Composition Diagnostics

Date

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Purpose

Fix the reviewer-revised research state for external verification. This version incorporates AC-level review concerns: endpoint leakage, syntax-holdout overclaiming, McNemar tests, semantic-control statistics, and the terminology change to a third-order signed permutation coherence test.

Track 1 fixed result

Delta vectors add reproducible information beyond target-only endpoints in the multiseed full-semantic setting. The syntax=1.0 result has now been directly ablated: y_only also reaches 1.0, so the syntax split is interpreted as endpoint/surface leakage rather than as geometry. BERT-large and DeBERTa-v3-base spot-checks both rank delta best under Linear SVC and logistic regression. UPAT is now reported as a hard-holdout boundary where delta is not reliably better than y_only.

Track 2 fixed result

Composition order matters. Semantic equivalence shifts noncommutativity distributions with strong statistical tests. The third-order result is now framed as signed permutation coherence, not a formal Jacobi identity. Decoder pairwise composition is now included. After multiple-testing correction, QMT is the only below-null triple that is stable across all five tested models.

Main conclusion

The evidence supports a local QMT signed-permutation cancellation effect across tested encoder models, while negation-heavy triples expose the boundary of the phenomenon. The current work is promising but not submission-ready without the listed blocker experiments.

Delta vs endpoint ablation with McNemar evidence

model	delta_mean	delta_std	y_only_mean	y_only_std	concat_mean	concat_std	x_only_mean	diff_delta_y	diff_delta_concat	max_mcnemar_p	all_mcnemar_sig_0_001
bert-base-uncased	0.9075	0.0117	0.8632	0.0167	0.8969	0.0185	0.1667	0.0443	0.0106	0.0000	1.0000
distilgpt2	0.8572	0.0125	0.7265	0.0094	0.7604	0.0107	0.1667	0.1307	0.0968	0.0000	1.0000
distilroberta-base	0.8810	0.0053	0.8306	0.0056	0.8417	0.0060	0.1667	0.0504	0.0394	0.0000	1.0000
gpt2	0.8332	0.0076	0.7496	0.0152	0.7765	0.0093	0.1667	0.0836	0.0567	0.0000	1.0000
roberta-base	0.8817	0.0090	0.8391	0.0072	0.8666	0.0089	0.1667	0.0426	0.0151	0.0000	1.0000

Syntax representation ablation: y_only solves the split

model	classifier	concat	delta	x_only	y_only
bert-base-uncased	linear_svc	1.0000	1.0000	0.1667	1.0000
bert-base-uncased	logreg	0.9904	0.9910	0.1667	0.9861
distilgpt2	linear_svc	1.0000	1.0000	0.1667	1.0000
distilgpt2	logreg	0.9976	1.0000	0.1667	0.9957
distilroberta-base	linear_svc	1.0000	1.0000	0.1667	1.0000
distilroberta-base	logreg	1.0000	1.0000	0.1667	1.0000
gpt2	linear_svc	1.0000	1.0000	0.1667	1.0000
gpt2	logreg	0.9979	0.9924	0.1667	0.9876
roberta-base	linear_svc	1.0000	1.0000	0.1667	1.0000
roberta-base	logreg	1.0000	1.0000	0.1667	1.0000

Layerwise/pooling syntax sanity check: top rows

classifier	accuracy	model	layer	pooling	representation	n_base	n_train	n_test
linear_svc	1.0000	bert-base-uncased	0.0000	mean	delta	200.0000	2400.0000	4800.0000
linear_svc	1.0000	bert-base-uncased	0.0000	mean	y_only	200.0000	2400.0000	4800.0000
linear_svc	1.0000	bert-base-uncased	2.0000	mean	y_only	200.0000	2400.0000	4800.0000
linear_svc	1.0000	bert-base-uncased	1.0000	cls	delta	200.0000	2400.0000	4800.0000
linear_svc	1.0000	bert-base-uncased	1.0000	mean	y_only	200.0000	2400.0000	4800.0000
linear_svc	1.0000	bert-base-uncased	1.0000	mean	delta	200.0000	2400.0000	4800.0000
linear_svc	1.0000	bert-base-uncased	2.0000	mean	delta	200.0000	2400.0000	4800.0000
linear_svc	1.0000	bert-base-uncased	4.0000	mean	delta	200.0000	2400.0000	4800.0000
linear_svc	1.0000	bert-base-uncased	4.0000	mean	y_only	200.0000	2400.0000	4800.0000
linear_svc	1.0000	bert-base-uncased	3.0000	cls	delta	200.0000	2400.0000	4800.0000
linear_svc	1.0000	bert-base-uncased	3.0000	mean	delta	200.0000	2400.0000	4800.0000
linear_svc	1.0000	bert-base-uncased	3.0000	mean	y_only	200.0000	2400.0000	4800.0000
linear_svc	1.0000	bert-base-uncased	2.0000	cls	y_only	200.0000	2400.0000	4800.0000
linear_svc	1.0000	bert-base-uncased	2.0000	cls	delta	200.0000	2400.0000	4800.0000
linear_svc	1.0000	bert-base-uncased	12.0000	mean	y_only	200.0000	2400.0000	4800.0000
linear_svc	1.0000	bert-base-uncased	7.0000	mean	delta	200.0000	2400.0000	4800.0000
linear_svc	1.0000	bert-base-uncased	6.0000	last_token	delta	200.0000	2400.0000	4800.0000
linear_svc	1.0000	bert-base-uncased	6.0000	cls	delta	200.0000	2400.0000	4800.0000
linear_svc	1.0000	bert-base-uncased	11.0000	mean	y_only	200.0000	2400.0000	4800.0000
linear_svc	1.0000	bert-base-uncased	11.0000	mean	delta	200.0000	2400.0000	4800.0000

DeBERTa-v3-small modern spot-check

model	classifier	concat	delta	x_only	y_only
microsoft/deberta-v3-small	linear_svc	0.8225	0.8708	0.1667	0.8042
microsoft/deberta-v3-small	logreg	0.8283	0.7958	0.1667	0.7700

BERT-large and DeBERTa-v3-base spot-checks

model	classifier	concat	delta	x_only	y_only
bert-large-uncased	linear_svc	0.8508	0.9025	0.1667	0.8375
bert-large-uncased	logreg	0.7600	0.8542	0.1667	0.7500
microsoft/deberta-v3-base	linear_svc	0.7758	0.8117	0.1667	0.7467
microsoft/deberta-v3-base	logreg	0.6942	0.7517	0.1667	0.7258

UPAT hard-holdout representation ablation

model	feature	acc	f1	baseline
bert-base-uncased	x_only	0.2000	0.0667	0.2000
bert-base-uncased	y_only	0.8250	0.8266	0.2000
bert-base-uncased	delta	0.7250	0.7253	0.2000
bert-base-uncased	concat_xy	0.8000	0.7974	0.2000
distilroberta-base	x_only	0.2000	0.0667	0.2000
distilroberta-base	y_only	0.6500	0.6105	0.2000
distilroberta-base	delta	0.7250	0.6530	0.2000
distilroberta-base	concat_xy	0.6250	0.5703	0.2000
roberta-base	x_only	0.2000	0.0667	0.2000
roberta-base	y_only	0.7250	0.7210	0.2000
roberta-base	delta	0.6750	0.6333	0.2000
roberta-base	concat_xy	0.7250	0.7185	0.2000
gpt2	x_only	0.2000	0.0667	0.2000
gpt2	y_only	0.4750	0.4358	0.2000
gpt2	delta	0.4250	0.3481	0.2000
gpt2	concat_xy	0.4750	0.4338	0.2000
distilgpt2	x_only	0.2000	0.0959	0.2000
distilgpt2	y_only	0.6000	0.5441	0.2000
distilgpt2	delta	0.6500	0.5953	0.2000
distilgpt2	concat_xy	0.5750	0.5169	0.2000

UPAT delta vs y_only McNemar tests

model	compare	delta_acc	y_only_acc	diff	b_delta_correct_y_wrong	c_delta_wrong_y_correct	p_value
bert-base-uncased	delta_vs_y_only	0.7250	0.8250	-0.1000	2.0000	6.0000	0.2891
distilroberta-base	delta_vs_y_only	0.7250	0.6500	0.0750	4.0000	1.0000	0.3750
roberta-base	delta_vs_y_only	0.6750	0.7250	-0.0500	7.0000	9.0000	0.8036
gpt2	delta_vs_y_only	0.4250	0.4750	-0.0500	7.0000	9.0000	0.8036
distilgpt2	delta_vs_y_only	0.6500	0.6000	0.0500	5.0000	3.0000	0.7266

Full-semantic pooling ablation

model	pooling	classifier	concat	delta	x_only	y_only
bert-base-uncased	cls	linear_svc	0.8269	0.8775	0.1667	0.7397
bert-base-uncased	cls	logreg	0.7031	0.7200	0.1667	0.6781
bert-base-uncased	last_token	linear_svc	0.7564	0.8514	0.1667	0.7508
bert-base-uncased	last_token	logreg	0.6792	0.7103	0.1667	0.6847
bert-base-uncased	mean	linear_svc	0.8850	0.8958	0.1667	0.8806
bert-base-uncased	mean	logreg	0.8725	0.8844	0.1667	0.8603
gpt2	cls	linear_svc	0.3683	0.3867	0.1667	0.3683
gpt2	cls	logreg	0.3422	0.3203	0.1667	0.3683
gpt2	last_token	linear_svc	0.7469	0.7564	0.1667	0.7200
gpt2	last_token	logreg	0.6453	0.6683	0.1667	0.6253
gpt2	mean	linear_svc	0.7822	0.8450	0.1667	0.7539
gpt2	mean	logreg	0.7803	0.7492	0.1667	0.7633
roberta-base	cls	linear_svc	0.8236	0.8736	0.1667	0.7911
roberta-base	cls	logreg	0.7944	0.8514	0.1667	0.7669
roberta-base	last_token	linear_svc	0.7656	0.7650	0.1667	0.7042
roberta-base	last_token	logreg	0.7056	0.6756	0.1667	0.6739
roberta-base	mean	linear_svc	0.8631	0.8786	0.1667	0.8431
roberta-base	mean	logreg	0.8150	0.8092	0.1667	0.8017

Confusion analysis: negation vs non-negation recall

source	model	classifier	group	mean_recall	mean_error_rate	n_classes
full_semantic	bert-base-uncased	linear_svc	negation	1.0000	0.0000	1.0000
full_semantic	bert-base-uncased	linear_svc	non_negation	0.8725	0.1275	5.0000
full_semantic	bert-base-uncased	logreg	negation	0.9862	0.0138	1.0000
full_semantic	bert-base-uncased	logreg	non_negation	0.8817	0.1182	5.0000
full_semantic	distilgpt2	linear_svc	negation	1.0000	0.0000	1.0000
full_semantic	distilgpt2	linear_svc	non_negation	0.8020	0.1980	5.0000
full_semantic	distilgpt2	logreg	negation	0.9725	0.0275	1.0000
full_semantic	distilgpt2	logreg	non_negation	0.7295	0.2705	5.0000
full_semantic	distilroberta-base	linear_svc	negation	1.0000	0.0000	1.0000
full_semantic	distilroberta-base	linear_svc	non_negation	0.8470	0.1530	5.0000
full_semantic	distilroberta-base	logreg	negation	1.0000	0.0000	1.0000
full_semantic	distilroberta-base	logreg	non_negation	0.8252	0.1747	5.0000
full_semantic	gpt2	linear_svc	negation	0.9600	0.0400	1.0000
full_semantic	gpt2	linear_svc	non_negation	0.7973	0.2027	5.0000
full_semantic	gpt2	logreg	negation	0.7312	0.2687	1.0000
full_semantic	gpt2	logreg	non_negation	0.7410	0.2590	5.0000
full_semantic	roberta-base	linear_svc	negation	0.9988	0.0012	1.0000
full_semantic	roberta-base	linear_svc	non_negation	0.8570	0.1430	5.0000
full_semantic	roberta-base	logreg	negation	0.9300	0.0700	1.0000
full_semantic	roberta-base	logreg	non_negation	0.7760	0.2240	5.0000

Third-order signed permutation summary with bootstrap CI

model	triple	mean_relative_jacobi_m	relative_jacobi_norm_ci95_	relative_jacobi_norm_ci95_	mean_jacobi_to_null_m	jacobi_to_null_mean_ratio_ci9	jacobi_to_null_mean_ratio_ci9	mean_null_percentile_smaller_is	n
bert-base-uncased	NMT	0.3048	0.3005	0.3090	0.7750	0.7669	0.7829	0.3522	100.0000
bert-base-uncased	NQM	0.2695	0.2660	0.2727	0.8628	0.8533	0.8720	0.3850	100.0000
bert-base-uncased	NQT	0.3719	0.3674	0.3763	1.1170	1.1082	1.1259	0.7005	100.0000
bert-base-uncased	QMT	0.2760	0.2712	0.2807	0.6833	0.6773	0.6892	0.1018	100.0000
distilroberta-base	NMT	0.3522	0.3485	0.3558	1.1338	1.1276	1.1404	0.7066	100.0000
distilroberta-base	NQM	0.2847	0.2809	0.2882	1.0821	1.0735	1.0914	0.7777	100.0000
distilroberta-base	NQT	0.2777	0.2752	0.2802	1.0803	1.0765	1.0841	0.6274	100.0000
distilroberta-base	QMT	0.1849	0.1832	0.1866	0.6311	0.6267	0.6356	0.0997	100.0000
roberta-base	NMT	0.4315	0.4255	0.4371	1.2532	1.2423	1.2637	0.8057	100.0000
roberta-base	NQM	0.3122	0.3086	0.3158	0.9799	0.9743	0.9853	0.6475	100.0000
roberta-base	NQT	0.3452	0.3413	0.3490	1.0805	1.0739	1.0871	0.7059	100.0000
roberta-base	QMT	0.2190	0.2160	0.2222	0.6229	0.6168	0.6287	0.1013	100.0000

Decoder signed permutation replication

model	triple	mean_jacobi_norm	mean_relative_jacobi_n	mean_null_relative_jacobi	jacobi_to_null_mean	null_percentile_smaller	n	relative_jacobi_norm_ci95	relative_jacobi_norm_ci95_	jacobi_to_null_mean_ratio_c	jacobi_to_null_mean_ratio_ci95
distilgpt2	NMT	10.3687	0.2276	0.2387	0.9507	0.5781	100.0000	0.2171	0.2388	0.9284	0.9722
distilgpt2	NQM	9.5684	0.2144	0.2540	0.8464	0.4061	100.0000	0.2056	0.2239	0.8249	0.8692
distilgpt2	NQT	15.8990	0.3385	0.2403	1.4202	0.8621	100.0000	0.3241	0.3537	1.3820	1.4555
distilgpt2	QMT	9.8263	0.2492	0.3179	0.7708	0.3207	100.0000	0.2320	0.2668	0.7446	0.7972
gpt2	NMT	31.1167	0.2894	0.2137	1.3348	0.7670	100.0000	0.2736	0.3065	1.2981	1.3723
gpt2	NQM	24.9467	0.2799	0.2315	1.1947	0.7302	100.0000	0.2642	0.2955	1.1675	1.2221
gpt2	NQT	16.4593	0.1562	0.1777	0.8666	0.4331	100.0000	0.1466	0.1658	0.8356	0.9004
gpt2	QMT	14.3721	0.1705	0.3167	0.5393	0.1852	100.0000	0.1621	0.1797	0.5192	0.5611

Signed permutation multiple-testing correction

model	triple	n	mean_ratio_to_null	bootstrap_p_ratio_less_than	direction	bonferroni_p	bh_fdr_p	passes_bonferroni_0_05	passes_bh_fdr_0_05
bert-base-uncased	NMT	100.0000	0.7750	0.0000	below_null	0.0010	0.0001	1.0000	1.0000
bert-base-uncased	NQM	100.0000	0.8628	0.0000	below_null	0.0010	0.0001	1.0000	1.0000
bert-base-uncased	QMT	100.0000	0.6833	0.0000	below_null	0.0010	0.0001	1.0000	1.0000
distilgpt2	NQM	100.0000	0.8464	0.0000	below_null	0.0010	0.0001	1.0000	1.0000
distilgpt2	QMT	100.0000	0.7708	0.0000	below_null	0.0010	0.0001	1.0000	1.0000
distilroberta-base	QMT	100.0000	0.6311	0.0000	below_null	0.0010	0.0001	1.0000	1.0000
gpt2	NQT	100.0000	0.8666	0.0000	below_null	0.0010	0.0001	1.0000	1.0000
gpt2	QMT	100.0000	0.5393	0.0000	below_null	0.0010	0.0001	1.0000	1.0000
roberta-base	NQM	100.0000	0.9799	0.0000	below_null	0.0010	0.0001	1.0000	1.0000
roberta-base	QMT	100.0000	0.6229	0.0000	below_null	0.0010	0.0001	1.0000	1.0000
distilgpt2	NMT	100.0000	0.9507	0.0001	below_null	0.0020	0.0002	1.0000	1.0000
bert-base-uncased	NQT	100.0000	1.1170	1.0000	above_null	1.0000	1.0000	0.0000	0.0000
distilgpt2	NQT	100.0000	1.4202	1.0000	above_null	1.0000	1.0000	0.0000	0.0000
distilroberta-base	NMT	100.0000	1.1338	1.0000	above_null	1.0000	1.0000	0.0000	0.0000
distilroberta-base	NQM	100.0000	1.0821	1.0000	above_null	1.0000	1.0000	0.0000	0.0000
distilroberta-base	NQT	100.0000	1.0803	1.0000	above_null	1.0000	1.0000	0.0000	0.0000
gpt2	NMT	100.0000	1.3348	1.0000	above_null	1.0000	1.0000	0.0000	0.0000
gpt2	NQM	100.0000	1.1947	1.0000	above_null	1.0000	1.0000	0.0000	0.0000
roberta-base	NMT	100.0000	1.2532	1.0000	above_null	1.0000	1.0000	0.0000	0.0000
roberta-base	NQT	100.0000	1.0805	1.0000	above_null	1.0000	1.0000	0.0000	0.0000

Decoder pairwise composition summary

model	pair	mean_commutator_nor	std_commutator_norm	mean_relative_commutato	std_relative_commutator_r	mean_cosine	std_cosine	mean_noncommutativit	std_noncommutativity	n
distilgpt2	MT_vs_TM	10.5809	1.8042	0.8545	0.2156	0.7111	0.1725	0.2889	0.1725	80.0000
distilgpt2	NM_vs_MN	3.7641	0.9414	0.3098	0.0893	0.9583	0.0283	0.0417	0.0283	80.0000
distilgpt2	NQ_vs_QN	5.7897	0.6043	0.4841	0.1058	0.8844	0.0472	0.1156	0.0472	80.0000
distilgpt2	NT_vs_TN	11.1703	1.8386	0.8664	0.1749	0.7046	0.1705	0.2954	0.1705	80.0000
distilgpt2	QM_vs_MQ	7.7162	0.9341	0.6474	0.1543	0.8039	0.0973	0.1961	0.0973	80.0000
distilgpt2	QT_vs_TQ	2.7566	0.6003	0.3249	0.1427	0.9571	0.0470	0.0429	0.0470	80.0000
gpt2	MT_vs_TM	11.3260	1.9708	0.3072	0.0694	0.9707	0.0080	0.0293	0.0080	80.0000
gpt2	NM_vs_MN	5.2897	1.1579	0.1894	0.0387	0.9901	0.0035	0.0099	0.0035	80.0000
gpt2	NQ_vs_QN	11.1700	1.5640	0.3855	0.0863	0.9337	0.0283	0.0663	0.0283	80.0000
gpt2	NT_vs_TN	23.9958	3.0734	0.7100	0.1186	0.9667	0.0305	0.0333	0.0305	80.0000
gpt2	QM_vs_MQ	13.3556	2.9359	0.5125	0.1313	0.8888	0.0563	0.1112	0.0563	80.0000
gpt2	QT_vs_TQ	3.9191	0.8893	0.2190	0.0846	0.9761	0.0196	0.0239	0.0196	80.0000

Semantic equivalence control

model	label	mean_noncommutativity	std_noncommutativity	mean_relative_commutator_st	std_relative_commutator_nor	mean_cosine	std_cosine	n
bert-base-uncased	equivalent	0.1130	0.0409	0.4818	0.0809	0.8870	0.0409	400.0000
bert-base-uncased	non_equivalent	0.3291	0.1134	0.8065	0.1326	0.6709	0.1134	400.0000
distilroberta-base	equivalent	0.1295	0.0483	0.5406	0.1277	0.8705	0.0483	400.0000
distilroberta-base	non_equivalent	0.2319	0.0381	0.6868	0.0560	0.7681	0.0381	400.0000
roberta-base	equivalent	0.1543	0.0630	0.5831	0.1304	0.8457	0.0630	400.0000
roberta-base	non_equivalent	0.2784	0.0439	0.7465	0.0597	0.7216	0.0439	400.0000

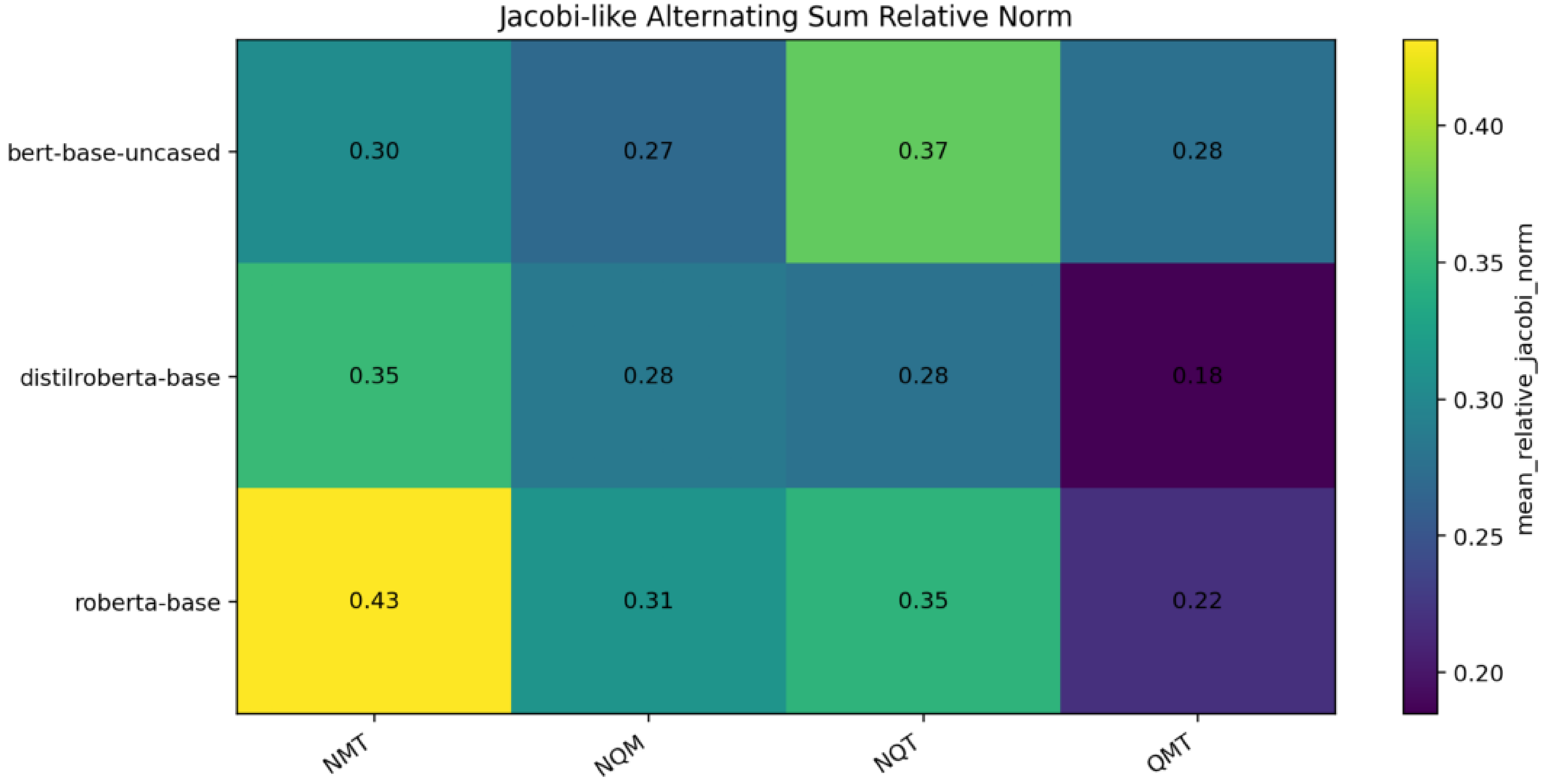
Semantic equivalence effect sizes

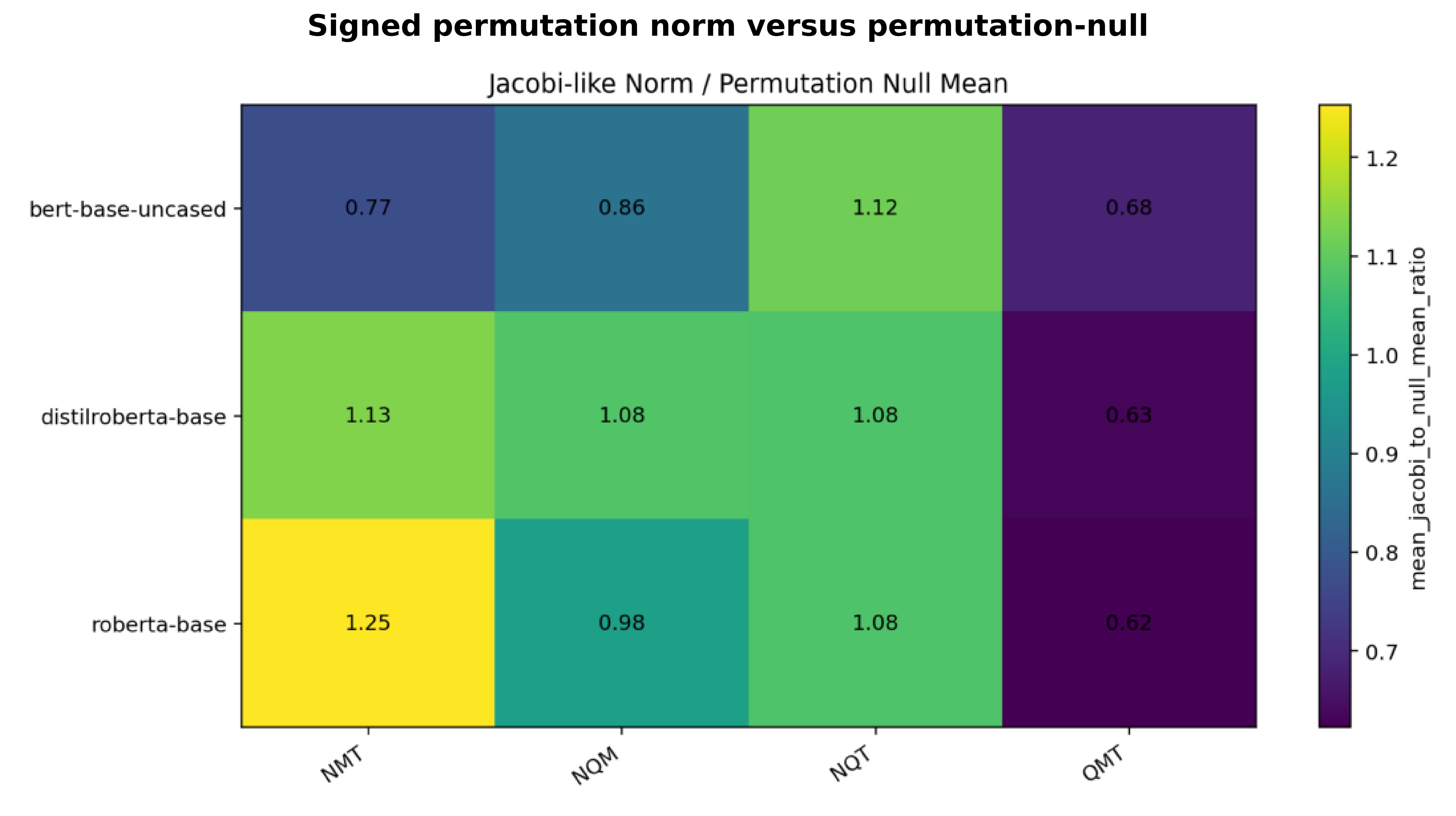
model	equiv_mean	non_equiv_mean	diff	cohens_d	equiv_std	non_equiv_std
bert-base-uncased	0.1130	0.3291	0.2161	2.5344	0.0409	0.1134
distilroberta-base	0.1295	0.2319	0.1024	2.3536	0.0483	0.0381
roberta-base	0.1543	0.2784	0.1241	2.2854	0.0630	0.0439

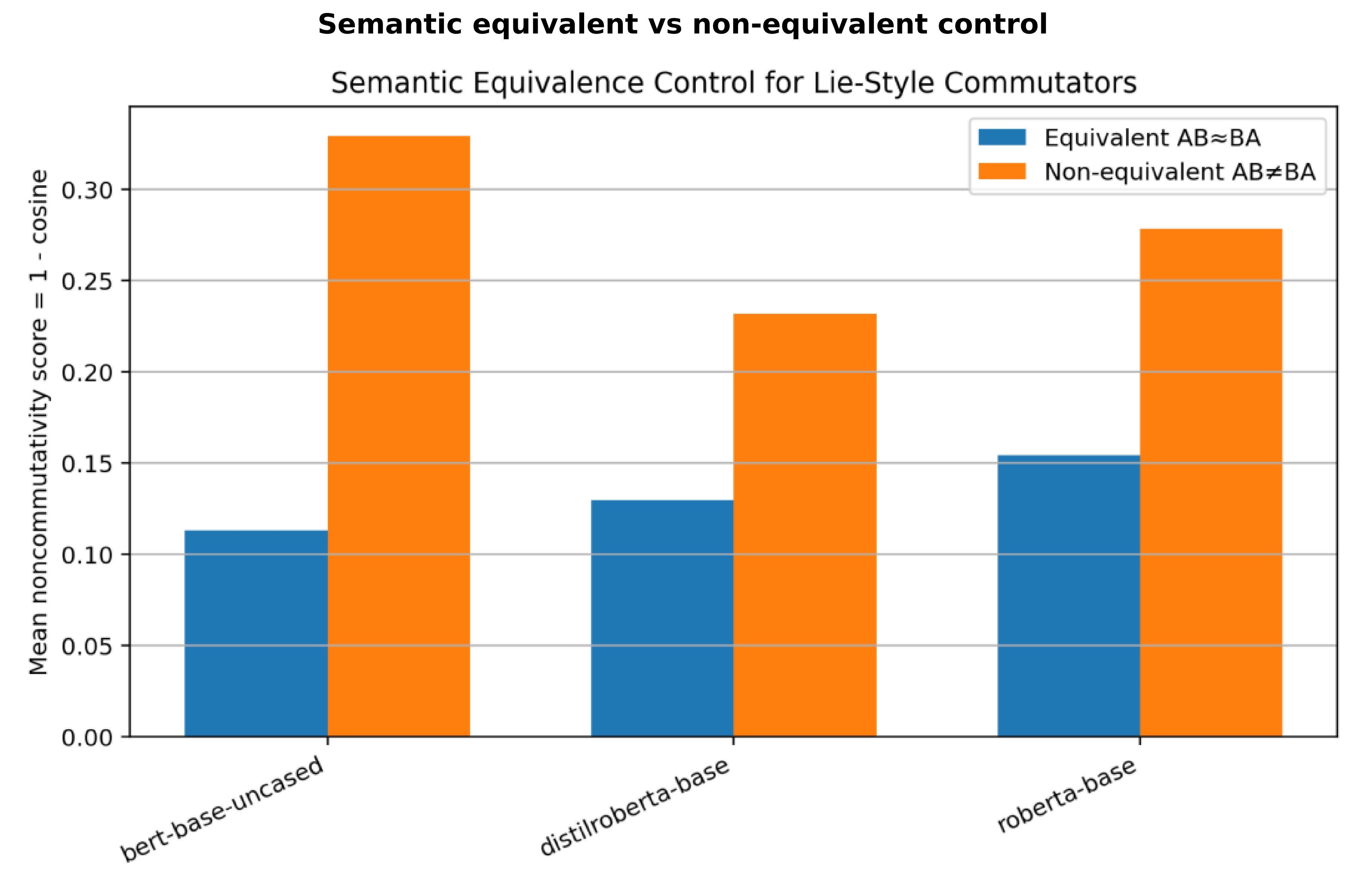
Composition summary: strongest noncommutativity rows

model	pair	mean_commutator_nor	std_commutator_norm	mean_relative_commutato	std_relative_commutator_n	mean_cosine	std_cosine	mean_noncommutativit	std_noncommutativity	n
bert-base-uncased	NT_vs_TN	5.1565	0.5832	0.9908	0.1066	0.5102	0.1009	0.4898	0.1009	80.0000
roberta-base	NQ_vs_QN	1.6422	0.0726	0.7846	0.0364	0.6928	0.0279	0.3072	0.0279	80.0000
bert-base-uncased	MT_vs_TM	4.2726	0.3075	0.7817	0.0954	0.6938	0.0727	0.3062	0.0727	80.0000
roberta-base	QM_vs_MQ	1.5649	0.0863	0.7714	0.0541	0.7032	0.0410	0.2968	0.0410	80.0000
distilroberta-base	MT_vs_TM	1.3504	0.0606	0.7683	0.0387	0.7124	0.0302	0.2876	0.0302	80.0000
bert-base-uncased	QM_vs_MQ	4.4793	0.2156	0.7341	0.0487	0.7340	0.0347	0.2660	0.0347	80.0000
roberta-base	MT_vs_TM	1.3956	0.0770	0.7215	0.0560	0.7411	0.0394	0.2589	0.0394	80.0000
roberta-base	NT_vs_TN	1.4185	0.1071	0.7172	0.0599	0.7429	0.0428	0.2571	0.0428	80.0000
bert-base-uncased	NQ_vs_QN	4.0903	0.2939	0.6970	0.0493	0.7598	0.0347	0.2402	0.0347	80.0000
distilroberta-base	NQ_vs_QN	1.4541	0.0644	0.6731	0.0384	0.7735	0.0257	0.2265	0.0257	80.0000
distilroberta-base	NT_vs_TN	1.3975	0.0725	0.6675	0.0342	0.7783	0.0232	0.2217	0.0232	80.0000
distilroberta-base	QM_vs_MQ	1.3672	0.0842	0.6613	0.0460	0.7926	0.0287	0.2074	0.0287	80.0000
bert-base-uncased	QT_vs_TQ	3.1001	0.1729	0.5463	0.0440	0.8513	0.0231	0.1487	0.0231	80.0000
roberta-base	QT_vs_TQ	0.7784	0.0701	0.3860	0.0342	0.9269	0.0128	0.0731	0.0128	80.0000
distilroberta-base	QT_vs_TQ	0.6580	0.0281	0.3422	0.0173	0.9415	0.0059	0.0585	0.0059	80.0000
bert-base-uncased	NM_vs_MN	1.2527	0.2368	0.2305	0.0538	0.9728	0.0125	0.0272	0.0125	80.0000
roberta-base	NM_vs_MN	0.3865	0.0403	0.2032	0.0248	0.9794	0.0050	0.0206	0.0050	80.0000
distilroberta-base	NM_vs_MN	0.3128	0.0212	0.1721	0.0201	0.9852	0.0036	0.0148	0.0036	80.0000

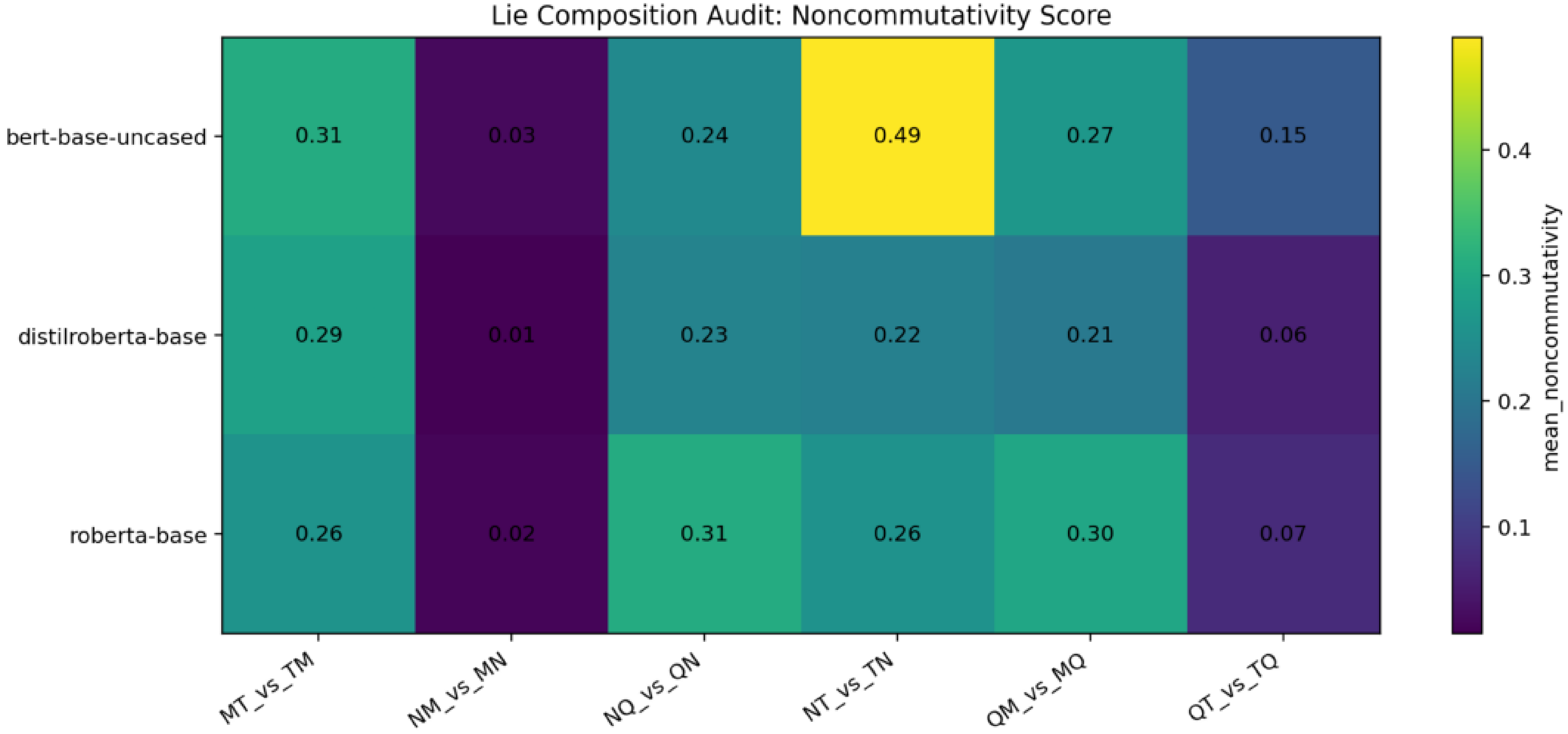
Signed permutation relative norm



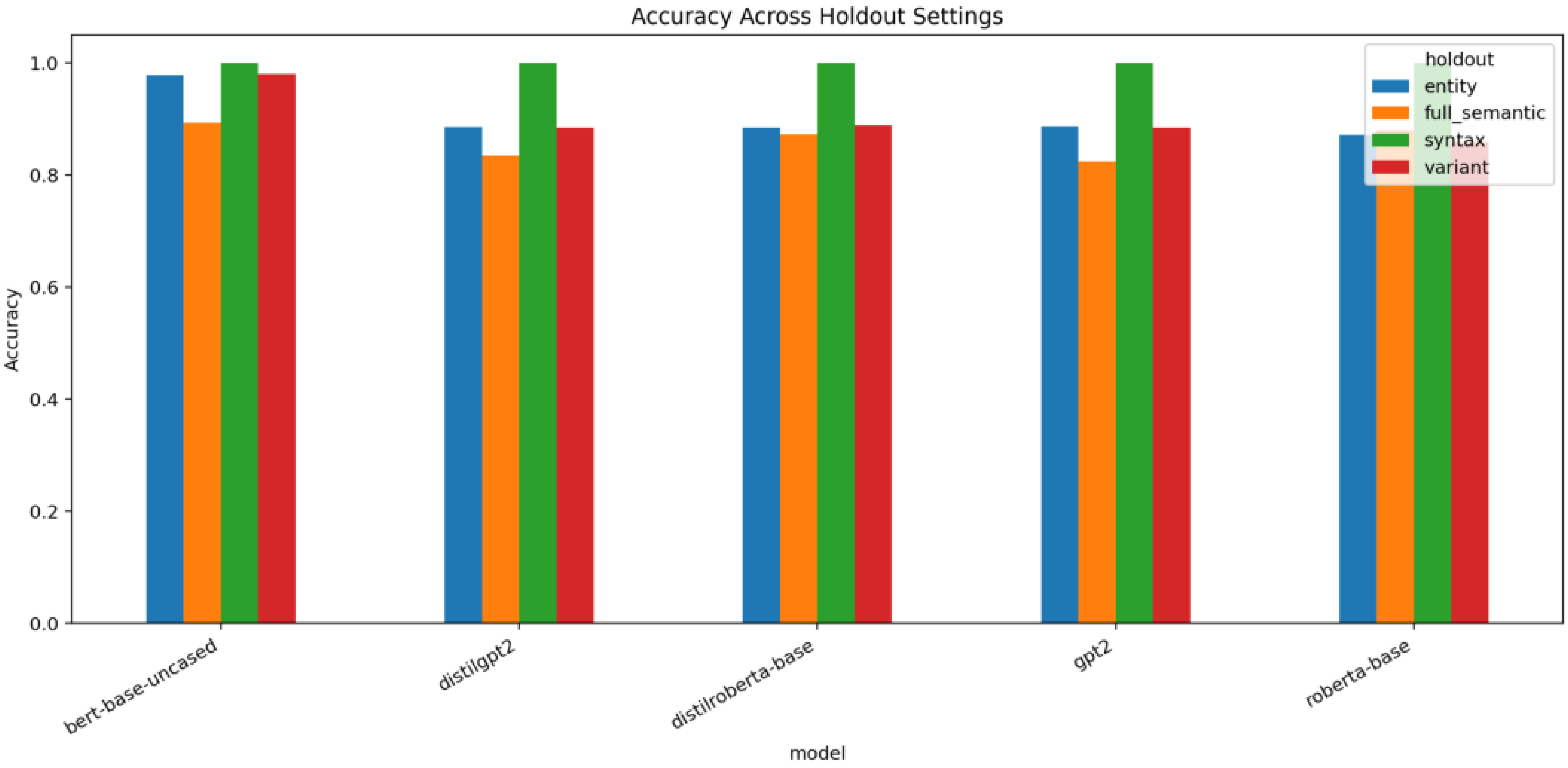

Signed permutation norm versus permutation-null
Jacobi-like Norm / Permutation Null Mean
bert-base-uncased
distilroberta-base
roberta-base
NMT
NQM
NQT
QMT
0.77
0.86
1.12
0.68
1.13
1.08
1.08
0.63
1.25
0.98
1.08
0.62
mean_jacobi_to_null_mean_ratio
0.7
0.8
0.9
1.0
1.1
1.2


Semantic equivalent vs non-equivalent control
Semantic Equivalence Control for Lie-Style Commutators
Equivalent AB≈BA
Non-equivalent AB≠BA
Mean noncommutativity score = 1 - cosine
0.00
0.05
0.10
0.15
0.20
0.25
0.30
bert-base-uncased
distilroberta-base
roberta-base

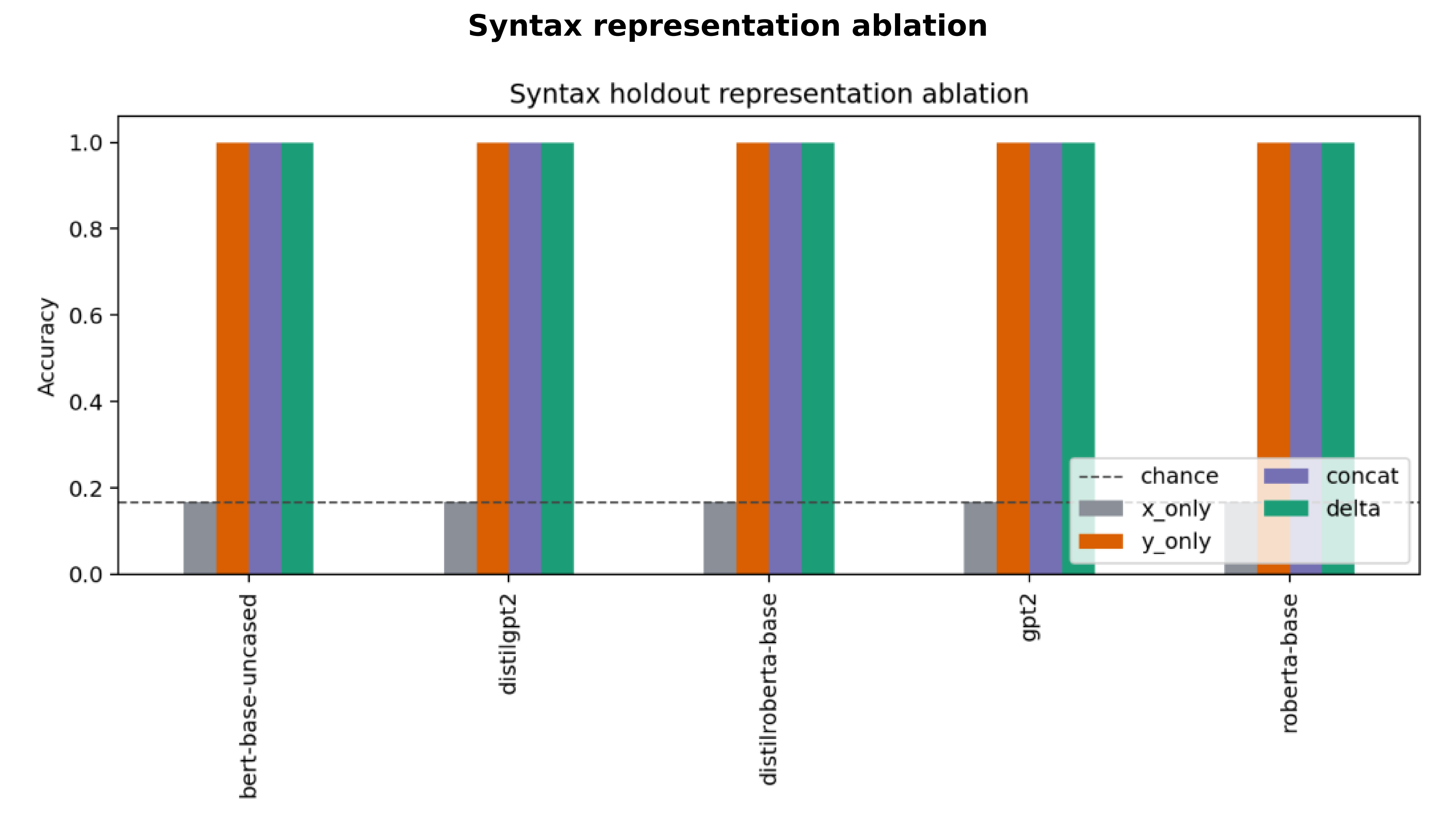
Composition noncommutativity heatmap



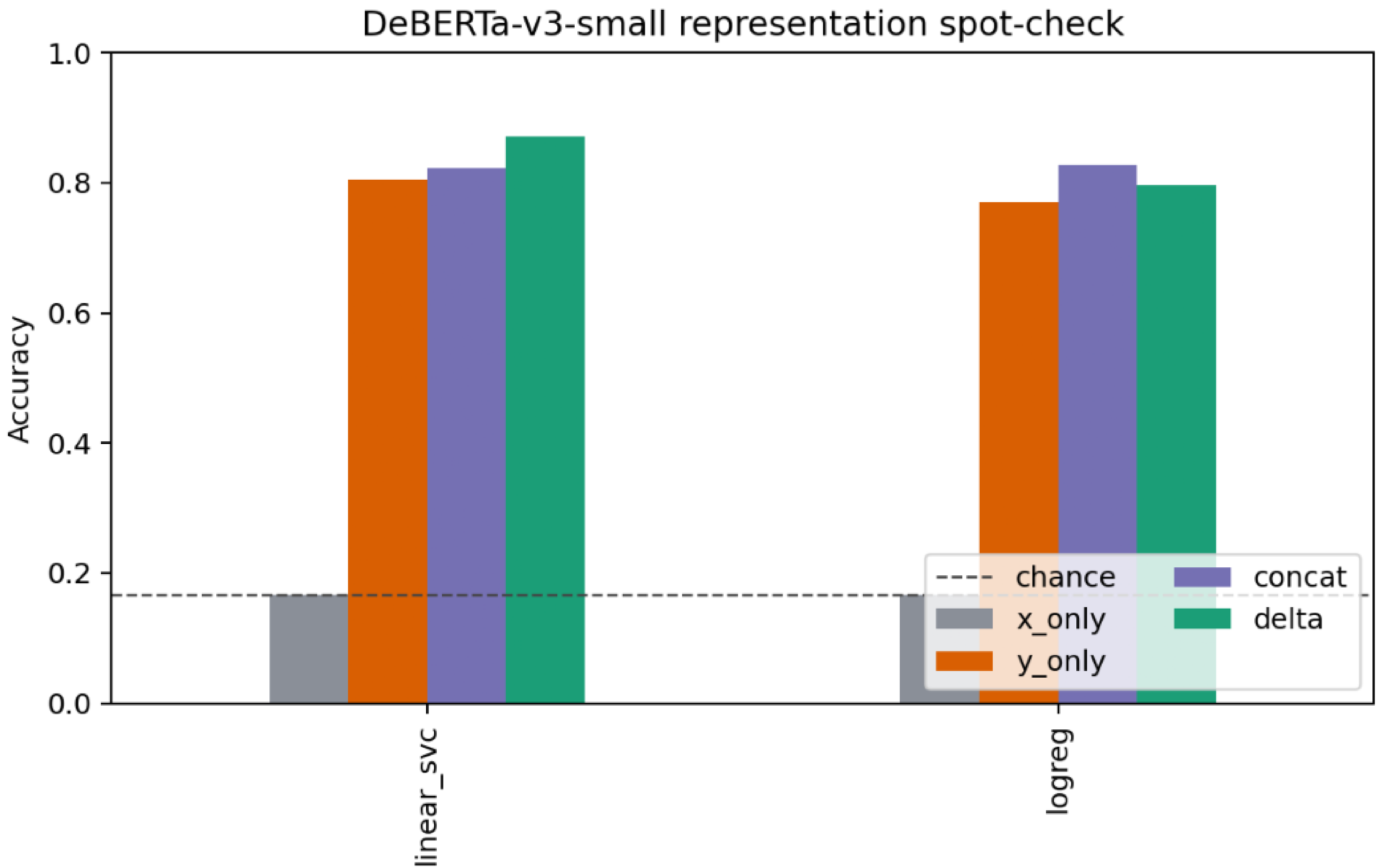
Original holdout accuracy comparison



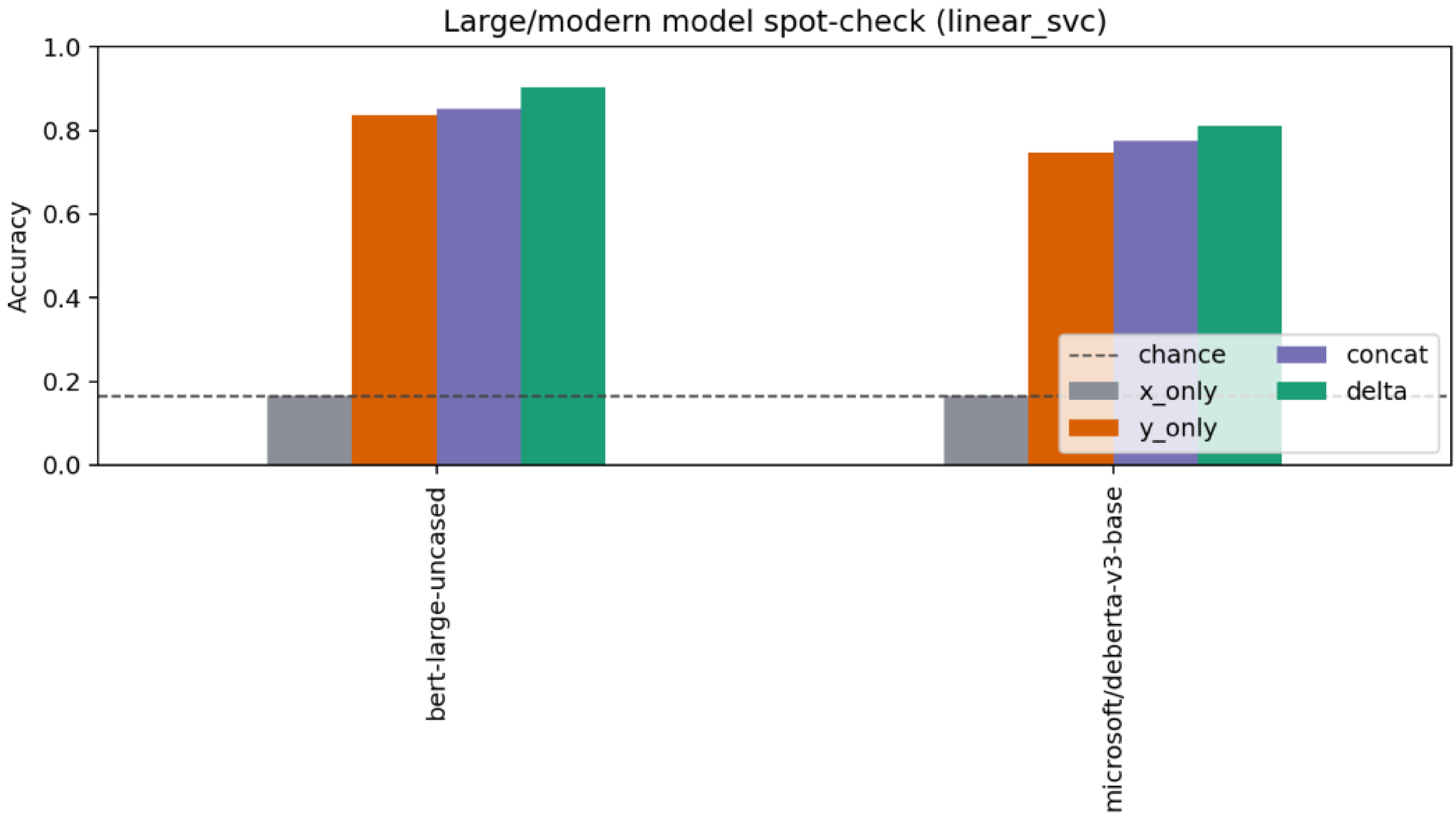
Original PCA class geometry


Syntax representation ablation
Syntax holdout representation ablation
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
chance
x_only
y_only
concat
delta
bert-base-uncased
distilgpt2
distilroberta-base
gpt2
roberta-base

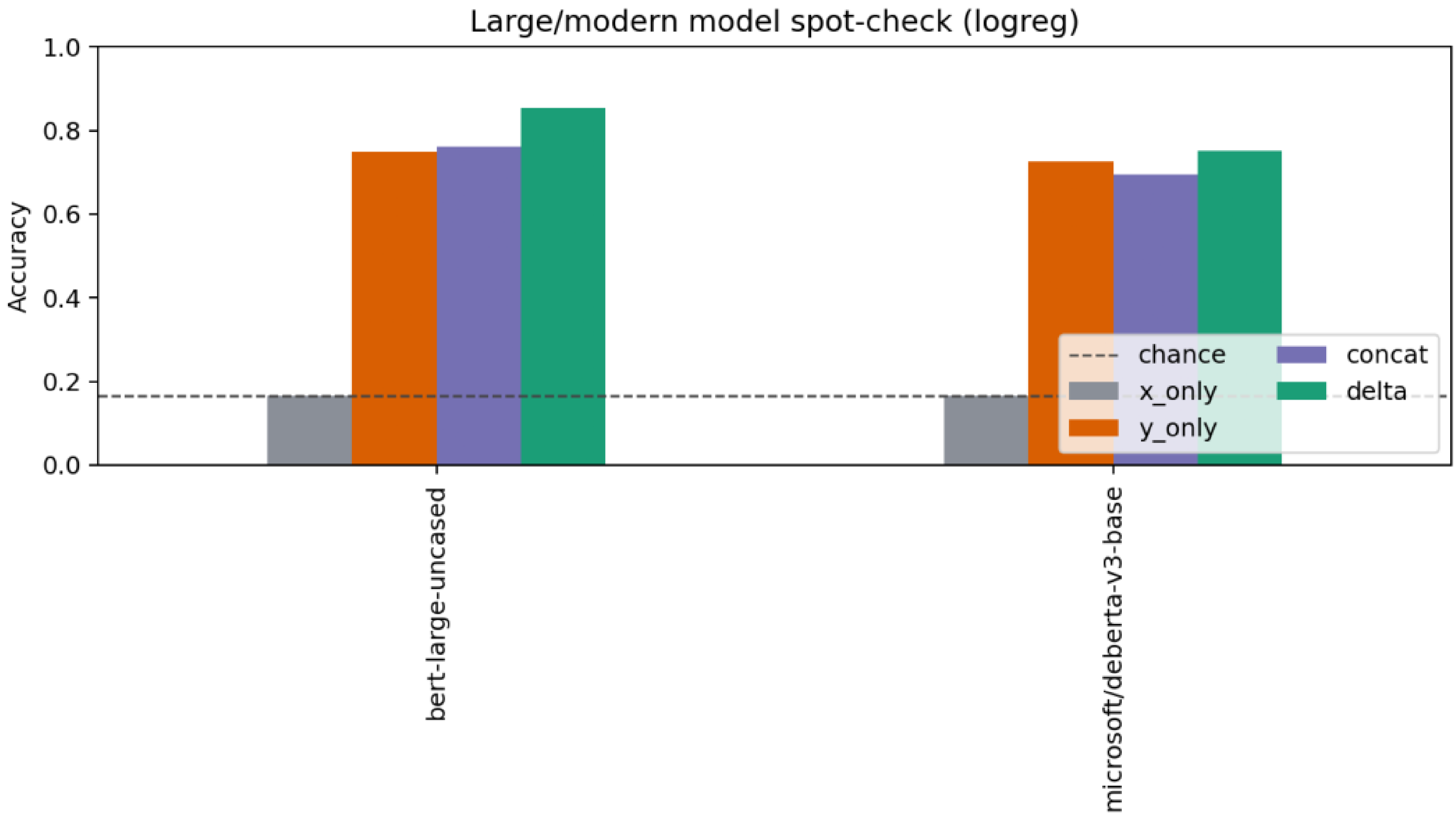
DeBERTa-v3-small spot-check



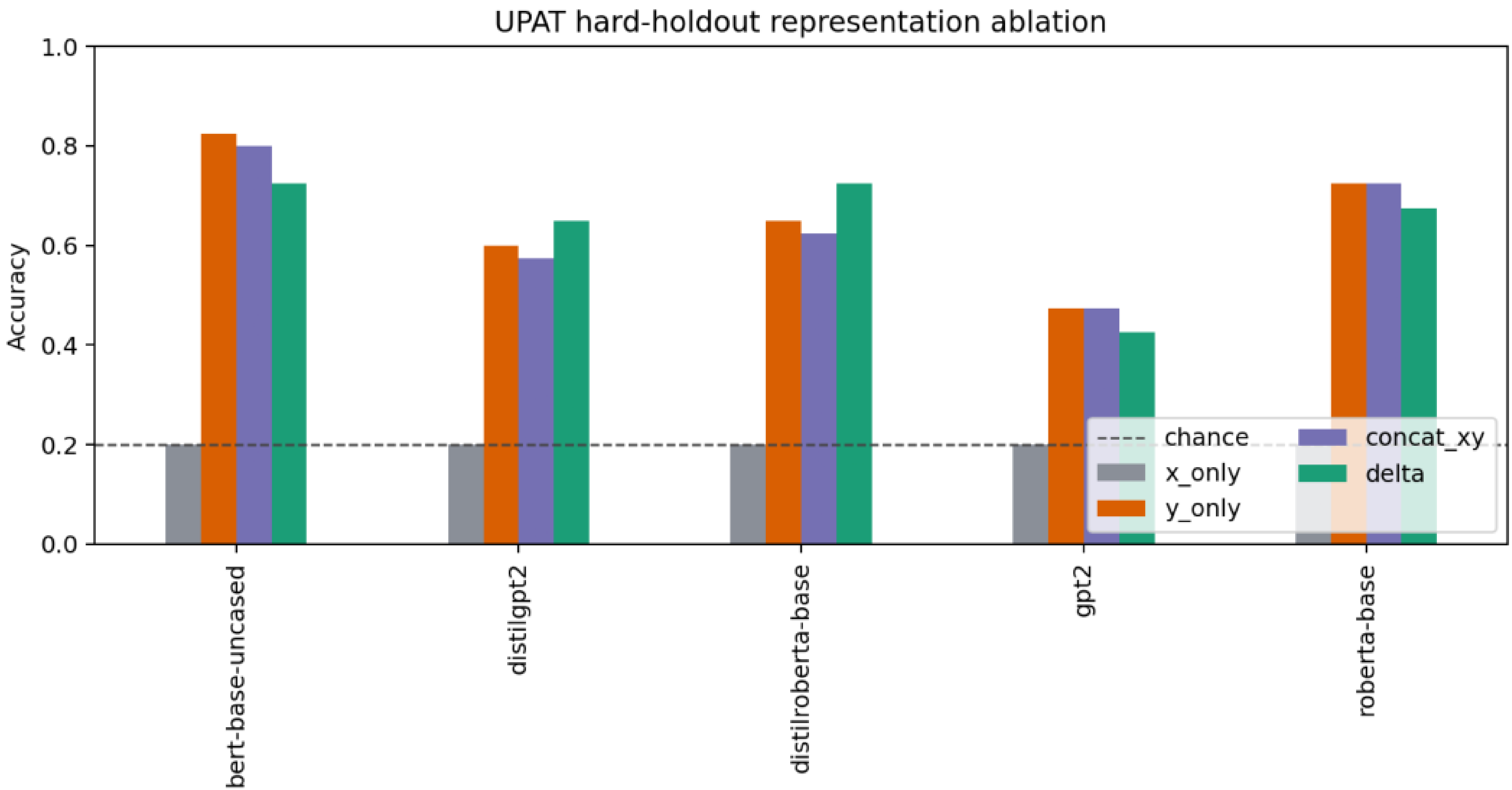
Large/modern spot-check: Linear SVC



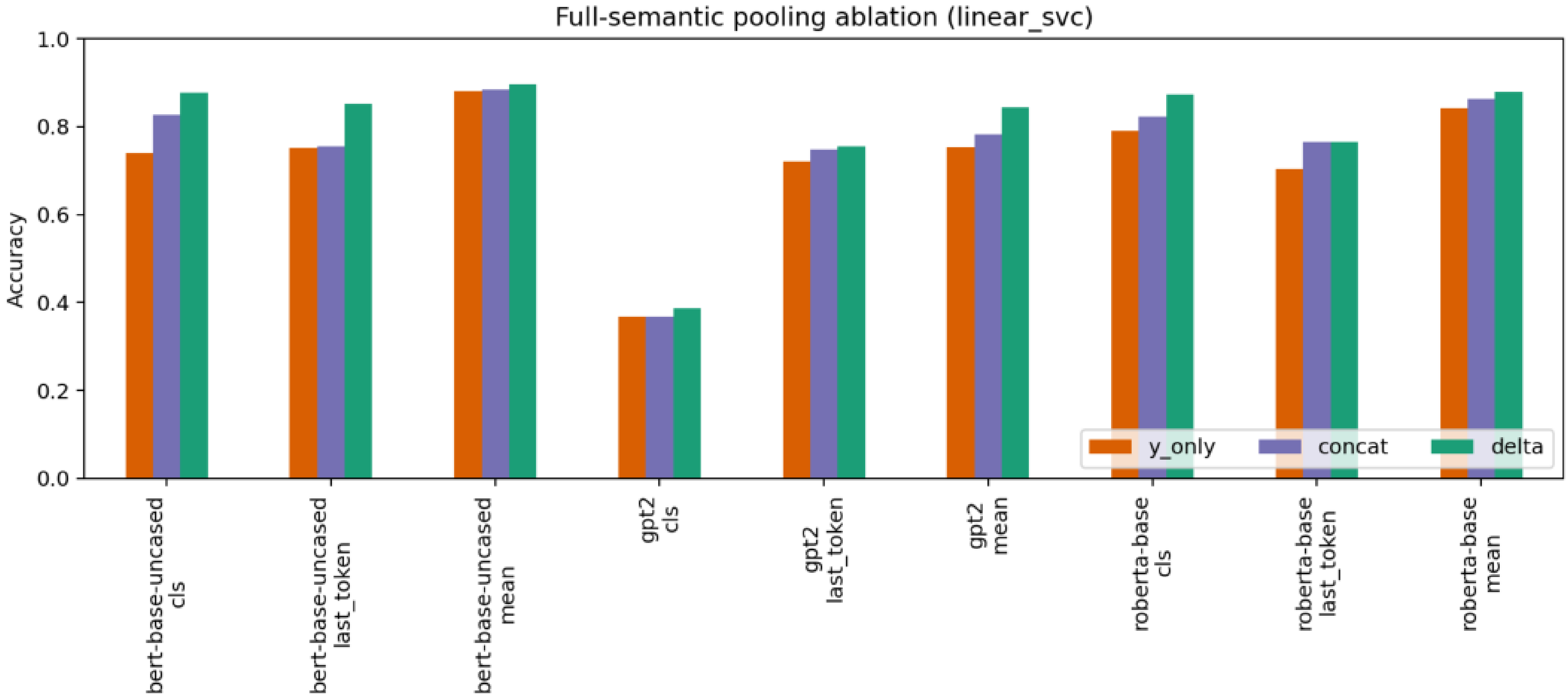
Large/modern spot-check: logistic regression



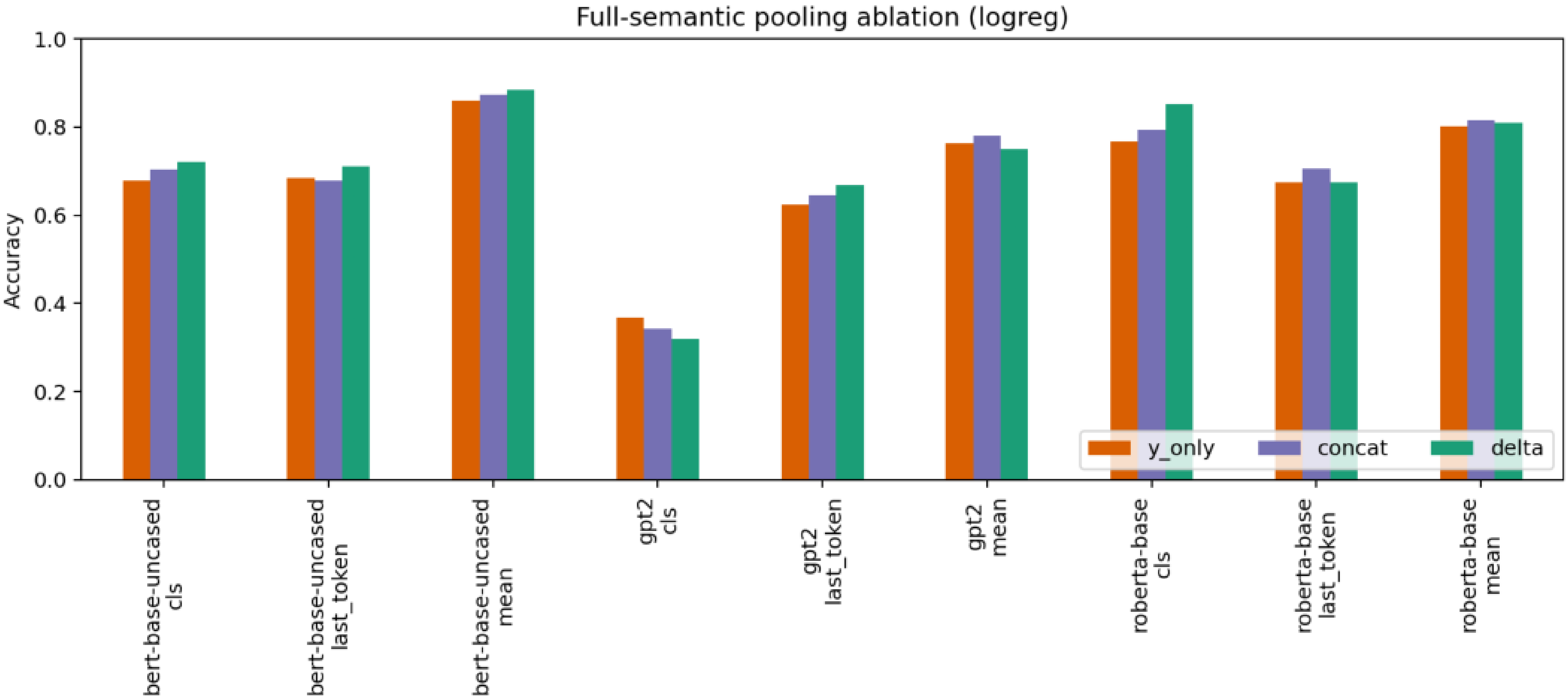
UPAT hard-holdout ablation

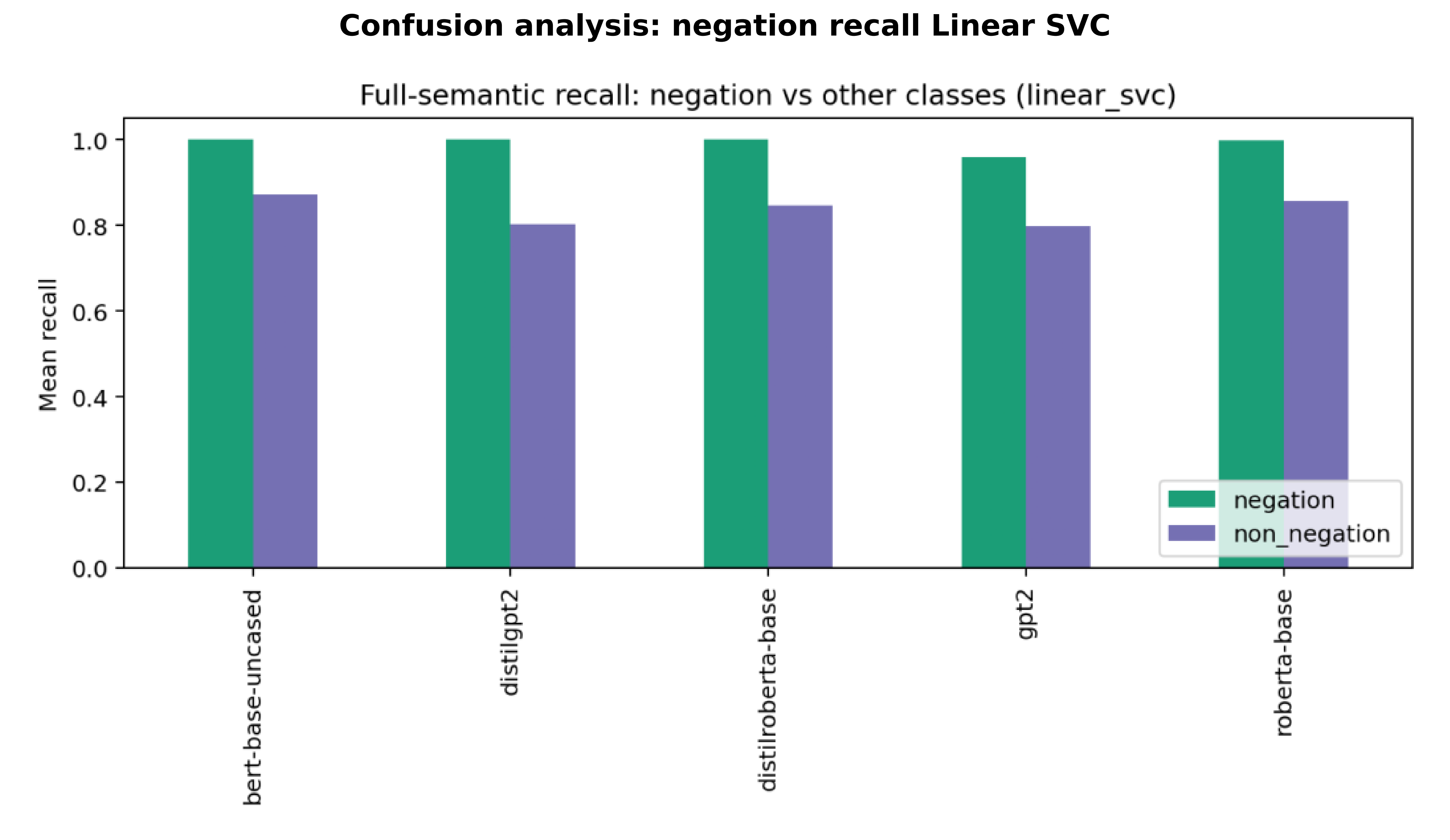


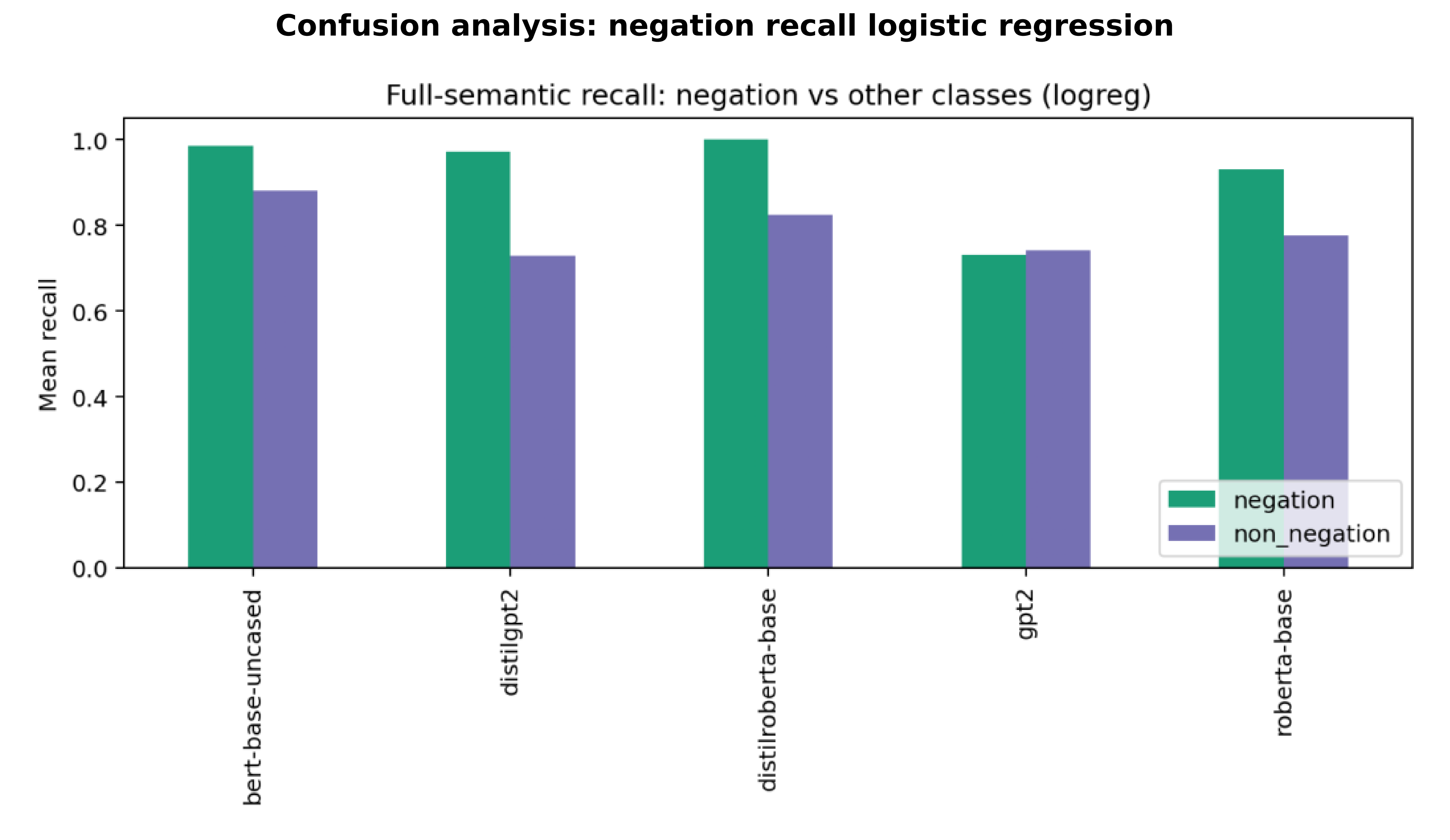
Full-semantic pooling ablation: Linear SVC



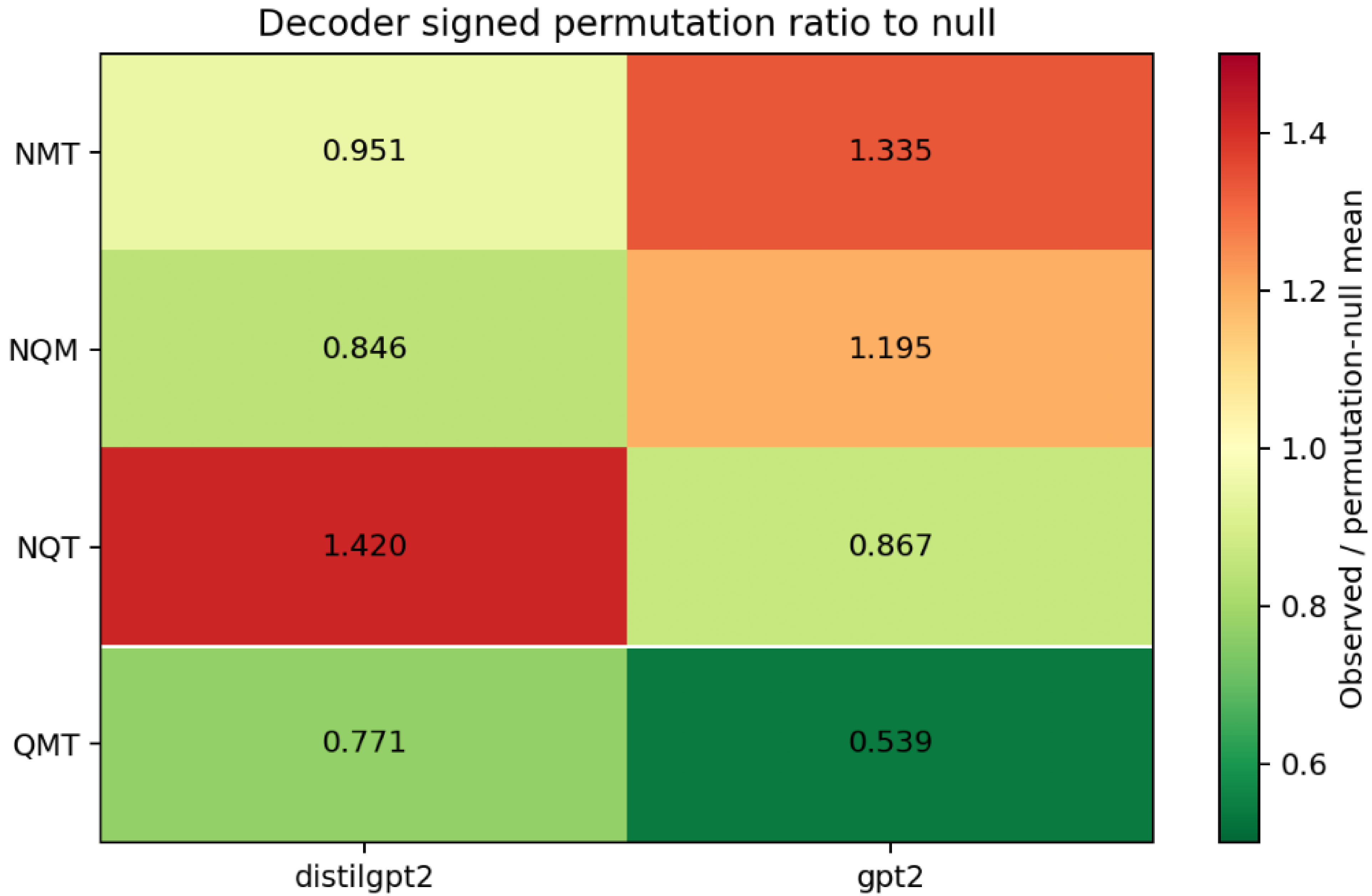
Full-semantic pooling ablation: logistic regression



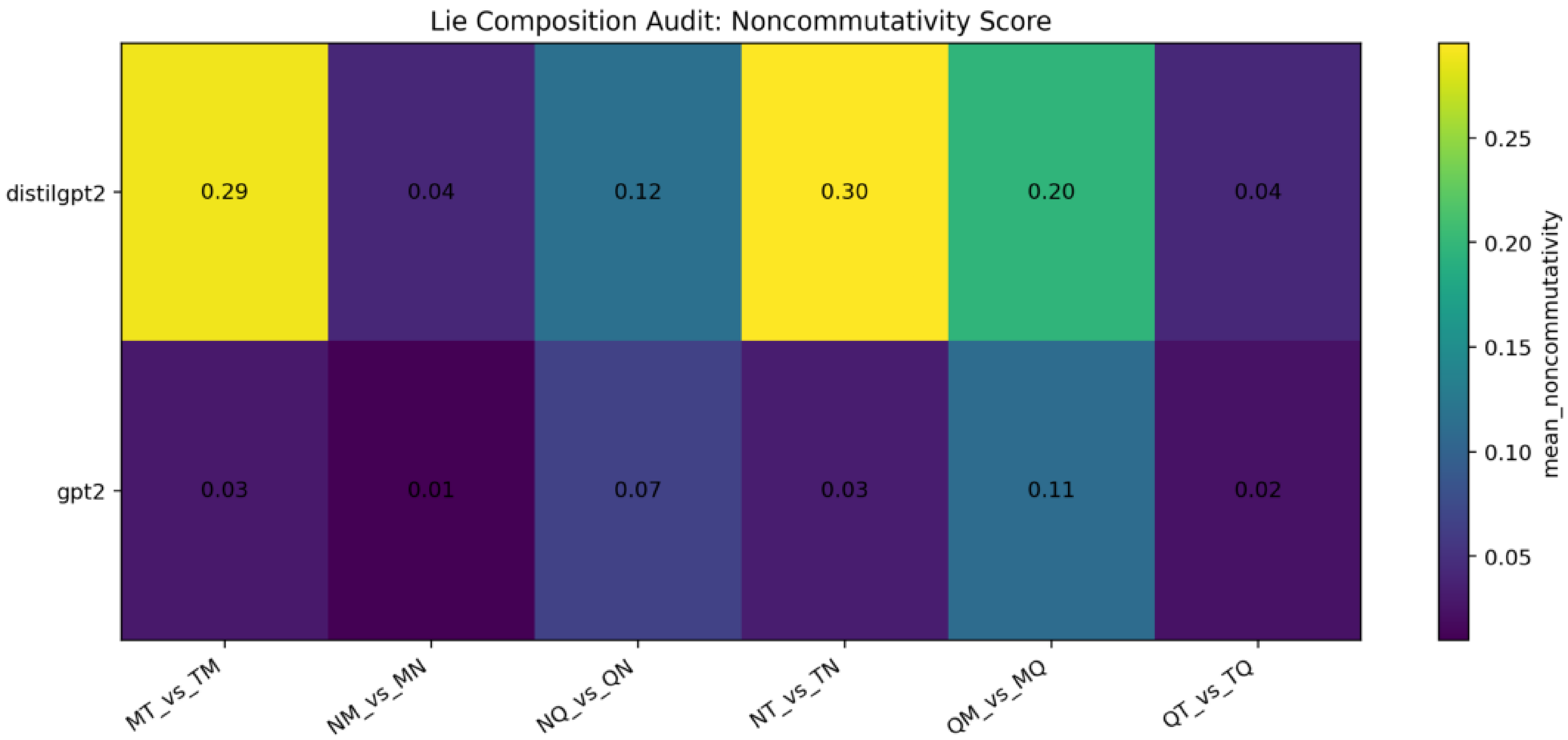

Confusion analysis: negation recall Linear SVC
Full-semantic recall: negation vs other classes (linear_svc)
Mean recall
1.0
0.8
0.6
0.4
0.2
0.0
negation
non_negation
bert-base-uncased
distilgpt2
distilroberta-base
gpt2
roberta-base


Confusion analysis: negation recall logistic regression
Full-semantic recall: negation vs other classes (logreg)
Mean recall
1.0
0.8
0.6
0.4
0.2
0.0
negation
non_negation
bert-base-uncased
distilgpt2
distilroberta-base
gpt2
roberta-base

Decoder signed permutation replication



Decoder pairwise composition



Audit notes for external verifier

Antisymmetry is not reported as evidence because the implementation defines [B,A] as the negative order of [A,B]. It is a tautological implementation check only.

A literal nested-commutator Jacobi expression over the same six endpoint vectors cancels algebraically. The reported third-order diagnostic is the non-tautological signed permutation composition sum ABC+BCA+CAB-ACB-CBA-BAC.

Duplicate endpoint templates were detected and removed before finalizing today's result. The final dataset has 400 Jacobi rows and zero duplicate endpoint sets.

New Track 1 result: syntax y_only=1.0 confirms target/surface leakage for the syntax split. New Track 1 spot-check: DeBERTa-v3-small supports delta superiority for Linear SVC but not for logistic regression. New Track 2 result: GPT-2 and DistilGPT-2 both keep QMT below permutation null, while negation triples remain mixed.

2026-06-13 update: after cache cleanup, BERT-large and DeBERTa-v3-base were both run successfully. Delta is best for both Linear SVC and logistic regression on both models.

Second 2026-06-13 update: UPAT is now explicitly reported as a hard-holdout boundary condition. Full-semantic pooling ablation, confusion/negation analysis, decoder pairwise composition, and signed-permutation multiple-testing correction are included.

Remaining blockers: resolve or expand UPAT, representation-ablation tables for every non-syntax holdout split, grammar-generated templates, endpoint-only controls for Track 2, composition layer/pooling ablations, and final bibliography formatting.

Round-3 Reviewer Response

Reviewer Response Round 3

Date: 2026-06-13

This note records the current response to the latest review of the research package.

Fixed in the Current Drafts

Antisymmetry

The Track 2 draft no longer presents antisymmetry as a discovery. It is now described as a tautological implementation check:

```text
[A,B] = delta_AB - delta_BA
[B,A] = delta_BA - delta_AB
```

Therefore `[A,B] = -[B,A]` follows for arbitrary vectors and does not provide evidence about transformer representations.

UPAT and y_only Confounding

The Track 1 draft now explicitly reports UPAT as a hard-holdout boundary result. UPAT sometimes favors `y_only` over `delta`, and no UPAT `delta` vs `y_only` McNemar comparison is significant. The claim is narrowed:

> Delta can add useful relational information, but endpoint-only features remain a serious confounder and can dominate under smaller, harder holdouts.

Layer-0 Syntax Result

The layer-0 syntax result is now framed as a red flag. Perfect separability at the embedding layer indicates token/form cues rather than deep semantic geometry.

Synthetic Lie Templates

The Track 2 draft now explicitly states that the current Lie-style composition dataset is synthetic and contains stable lexical markers. The result is a controlled diagnostic, not a general natural-language operator algebra.

Mixed Signed-Permutation Results

The Track 2 draft now explains that several triples are below null in some models, while others are above null. `QMT` is promoted only because it is the only tested triple that is below null across all five models after table-level correction.

Still Required Before Submission

1. Add a Procrustes alignment null baseline for UPAT:
   - random-label or random-pairing alignment
   - report null distribution for F1 gain
2. Increase shuffle controls from 100 permutations to at least 1000 before reporting precise p-values.
3. Add null baselines for pairwise commutator norms `||[A,B]||`.
4. Replace hand-written Lie templates with grammar-generated paraphrase templates.
5. Run endpoint-only controls for composition endpoints.
6. Convert large/modern Track 1 spot-checks into multiseed runs if Track 1 is promoted to submission.

Current Submission Readiness

The package is now stronger as a research record, but it should not be submitted as a main-track paper yet. The best current framing is:

- Track 1: a representation-geometry study with clear endpoint confounders and a hard-holdout boundary.
- Track 2: a diagnostic study of local signed-permutation coherence, with `QMT` as the only cross-model stable triple and negation as a composition instability.

Updated Research Roadmap

Research Program: From Linguistic Transformation Vectors to Composition Diagnostics

Core question

Can linguistic transformations be represented not only as separable classes in transformer embedding spaces, but as reusable geometric operations with meaningful composition structure?

The project currently has two paper tracks.

Strategic narrative

The strongest long-term story is broader than either current draft:

> Transformers may encode linguistic operators as geometric objects in a low-dimensional, partially universal transformation subspace, and this subspace may be transferable across architectures and languages.

Current results are not enough to claim this yet. The Procrustes transfer numbers are promising, but they remain exploratory until null baselines are added. The roadmap below treats cross-model and cross-lingual transfer as the next central hypothesis, not as a finished conclusion.

Track 1: Geometric transformation vectors

Working title:

Geometric Transformation Vectors in Transformer Embedding Spaces

Central claim:

Transformer embedding spaces contain displacement directions that encode linguistic transformations. These directions are recoverable across multiple models and holdout regimes.

Current evidence:

- Delta vectors classify transformations well across BERT, RoBERTa, DistilRoBERTa, GPT-2, and DistilGPT-2.
- Syntax holdouts reach `1.0` accuracy across tested models, but the new representation ablation shows `y_only=1.0` as well. This is now interpreted as endpoint/surface leakage, not as deep generalization.
- Full semantic holdouts remain above chance, with accuracies around `0.82-0.89`.
- Entity and variant holdouts remain strong, especially for BERT-family models.

- Modern/larger spot-checks now support the main delta advantage:
  - BERT-large Linear SVC: `delta=0.903`, `concat=0.851`, `y_only=0.838`
  - BERT-large logistic regression: `delta=0.854`, `concat=0.760`, `y_only=0.750`
  - DeBERTa-v3-base Linear SVC: `delta=0.812`, `concat=0.776`, `y_only=0.747`
  - DeBERTa-v3-base logistic regression: `delta=0.752`, `concat=0.694`, `y_only=0.726`
- UPAT hard-holdout is a boundary result: `delta` is worse than `y_only` for BERT, RoBERTa, and GPT-2, and no UPAT delta-vs-y McNemar test is significant.
- Full-semantic pooling ablation now supports mean pooling as a defensible main choice, while showing that the decoder effect is partly pooling-dependent.

Scientific status:

This is the more mature paper. It can be written as a representation-geometry result with careful controls against template memorization.

Main risk:

Some signal may come from target-sentence artifacts rather than pure transformation geometry. The concrete high-risk case is now confirmed: `syntax=1.0` reproduces under `y_only`, so it must be interpreted as a target/surface artifact unless future controls overturn that. UPAT also shows that `delta > y_only` is not universal under small hard-holdout regimes. The paper needs strong source-only, target-only, delta-only, and paraphrase controls.

Track 2: Signed permutation coherence for linguistic operators

Working title:

Third-Order Signed Permutation Coherence for Linguistic Transformations in Transformer Embedding Spaces

Central claim:

Some linguistic transformations behave like locally noncommuting operations in embedding space. Certain triples show third-order signed permutation cancellation stronger than permutation-null baselines, but the evidence supports local composition structure rather than a global Lie algebra.

Current evidence:

- Composition order matters: `AB` and `BA` often differ substantially.
- Semantic equivalence controls show lower noncommutativity for equivalent pairs than for non-equivalent pairs.
- Antisymmetry is not treated as evidence. It is a tautological implementation check because `[A,B] = delta_AB - delta_BA` implies `[B,A] = -[A,B]`.
- The non-tautological third-order signed permutation diagnostic is:

```text
S(A,B,C) = ABC + BCA + CAB - ACB - CBA - BAC
```

- The `QMT` triple shows robust cancellation across tested models:
  - BERT: ratio to permutation null `0.683`, CI `[0.677, 0.689]`
  - DistilRoBERTa: ratio `0.631`, CI `[0.627, 0.636]`
  - RoBERTa: ratio `0.623`, CI `[0.617, 0.629]`
- Decoder replication is now partially complete:
  - GPT-2 `QMT`: ratio `0.539`, CI `[0.519, 0.561]`
  - DistilGPT-2 `QMT`: ratio `0.771`, CI `[0.745, 0.797]`
  - Negation-containing triples remain mixed and model-dependent.
- Decoder pairwise composition summaries are now added, not only decoder third-order summaries.
- Multiple-testing correction over `4 triples x 5 models` supports the narrowed claim: `QMT` is the only below-null triple passing across all five tested models.

Scientific status:

This is promising but more fragile. It should be framed as a null-controlled diagnostic, not as proof of a Lie algebra or as a formal Jacobi identity.

Main risk:

Hand-written templates may induce or suppress cancellation. GPT-2/DistilGPT-2 now support `QMT` below-null cancellation, but negation behaves inconsistently across architectures. The current Lie-style templates still contain stable lexical markers, so the next experiments need grammar-driven templates, endpoint-only controls, commutator null baselines, and a focused negation analysis before expanding the operator set.

Track 3: Cross-model transformation transfer

Working title:

Universal Transformation Subspaces Across Transformer Architectures

Central hypothesis:

Transformation geometry may be partially model-independent after low-dimensional alignment. A classifier trained on transformation deltas in one model may transfer to another model after Procrustes alignment.

Promising exploratory evidence:

- BERT to `all-mpnet-base-v2`: raw F1 around `0.15`, aligned F1 around `0.857`.
- BERT to `all-MiniLM-L6-v2`: raw F1 around `0.17`, aligned F1 around `0.754`.
- UPAT-large cross-model results show large aligned-transfer gains across several architectures.

Current status:

This could become the strongest paper if it survives null controls. It is not yet promoted because Procrustes has many degrees of freedom and can overfit small alignment sets.

Required gates before promotion:

1. Random-label or random-pairing Procrustes null baseline.
2. Alignment-size curve with held-out evaluation.
3. At least 1000 shuffle/permutation runs for precise p-values.
4. Reverse-direction transfer tests, e.g. small model to large model and large model to small model.
5. Stronger architectures if feasible, such as Llama/Mistral-class embedding spaces or high-quality sentence encoders.

Track 4: Transformation vectors as editors

Working title:

From Transformation Geometry to Controllable Linguistic Editing

Central hypothesis:

If transformation directions are real, they should not only classify transformations; they should also steer generation or hidden states toward those transformations.

Minimal experiment:

1. Use GPT-2 as the first testbed.
2. Compute a centroid delta for a transformation such as negation or question formation.
3. Inject the vector into the residual stream during generation.
4. Measure whether outputs systematically acquire the target transformation.
5. Compare against random-vector, norm-matched, and wrong-class controls.

Scientific payoff:

This would move the project from descriptive geometry to causal intervention. A positive result would connect the work to representation steering and controllable generation.

Track 5: Cross-lingual transformation geometry

Working title:

Language-Invariant Geometry of Linguistic Transformations

Central hypothesis:

If transformation geometry is universal rather than English-template-specific, aligned multilingual models should show comparable transformation subspaces across languages.

Minimal experiment:

1. Use mBERT or XLM-R.
2. Build parallel transformation pairs in English and one or more non-English languages.
3. Compare delta geometry within each language.
4. Align language-specific transformation spaces with Procrustes.
5. Test whether classifiers or centroids transfer across languages.

Scientific payoff:

This is the cleanest answer to the criticism that the current effects may be English grammar or template artifacts.

Track 6: Dimensionality of transformation manifolds

Working title:

Effective Dimensionality of Linguistic Transformation Subspaces

Central hypothesis:

Each transformation may occupy a low-dimensional subspace rather than a single direction or the full embedding space.

Measurements:

- PCA spectra of per-class delta matrices.
- Participation ratio of singular values.
- Classification accuracy as a function of retained dimensions.
- Sample-complexity curves for learning each transformation.

Scientific payoff:

This would quantify the complexity of each linguistic operator and explain why capacity curves keep improving with more examples.

Iterative logic

The research direction is:

1. Show transformations are encoded as recoverable displacement vectors.
2. Test whether those directions compose nontrivially.
3. Separate semantic order effects from wording artifacts.
4. Test local third-order signed permutation structure with null baselines.
5. Expand from hand-written probes to systematic grammar-generated probes.
6. Decide which claims are strong enough for publication.

Common methodology standards

Every promoted claim should have at least one control:

- holdout split
- source-only and target-only baseline
- permutation/null baseline
- semantic equivalence control
- bootstrap confidence interval
- cross-model replication
- template de-duplication check

Negative results remain part of the research record.

Next research plan

Short term

1. Close methodology blockers required for any submission:
   - remove antisymmetry from the evidence narrative; already done in the draft, keep it that way
   - add Procrustes null baselines
   - increase shuffle/permutation controls to at least 1000, ideally 5000 for final numbers
   - add commutator norm null baselines
2. Resolve UPAT:
   - either expand UPAT and match train sizes against the main dataset
   - or keep it explicitly as a hard negative control and narrow Track 1 claims
3. Add null baselines for exploratory UPAT alignment:
   - random-label or random-pairing Procrustes null
   - at least 1000 shuffle permutations instead of 100
4. Test cross-model transformation transfer as the likely central future paper:
   - reverse-direction transfer
   - alignment-size curve

   - random-label/null alignment controls
5. Treat the syntax holdout as resolved for the current draft: `y_only=1.0` and layer-0 `1.0` mean the `syntax=1.0` result is a target/surface artifact unless a future redesigned split proves otherwise.
6. Convert large/modern Track 1 spot-checks into multiseed runs if Track 1 is promoted to submission.
7. Build a grammar-driven template generator for `N,Q,M,T` that produces many paraphrases without duplicate endpoints.
8. Add automated dataset validation:
   - no duplicate endpoint strings within a composition tuple
   - balanced subjects/actions
   - controlled lexical overlap
9. Add commutator null baselines for `||[A,B]||` using random or label-shuffled operations with matched norms.
10. Re-run composition and signed-permutation diagnostics on generated templates.
11. Add target-only and endpoint-only baselines for the Lie-style paper.
12. Regenerate a dated PDF packet after every major run.

Medium term

1. Run the GPT-2 steering-vector experiment for one transformation, starting with negation or question formation.
2. Run the cross-lingual mBERT/XLM-R transformation-transfer experiment.
3. Measure effective dimensionality of transformation subspaces:
   - participation ratio
   - PCA retention curves
   - accuracy versus retained dimensions
4. Run layerwise and pooling ablations before expanding the operator set. If final-layer mean pooling is not the best representation, the expanded operator experiments should use the validated representation instead.
5. Add more linguistic operators:
   - passive voice
   - modality strength
   - evidentiality
   - aspect
   - conditionality
   - quantifier changes
6. Add paraphrase robustness with semantically equivalent endpoint variants.

Paper milestones

1. Track 1 is arXiv-ready only when this checklist is complete:
   - syntax holdout breakdown is in the draft and interpreted as endpoint leakage
   - McNemar tests are reported in the text

- multiseed standard deviations are reported
- at least one prior-work baseline or comparison is written up, such as task vectors or function vectors
- at least one model outside the original five is included as a spot-check; currently satisfied by BERT-large and DeBERTa-v3-base
- UPAT hard-holdout is either resolved experimentally or explicitly reported as a negative boundary condition

2. Keep Track 2 as a diagnostics paper until grammar-generated templates and endpoint-only controls succeed.
3. Promote Track 3 to the main paper only if Procrustes transfer survives null baselines. If it does, the main narrative becomes cross-architecture universal transformation subspaces.
4. If steering works, write Track 4 as an intervention/controllable-generation paper.
5. If cross-lingual transfer works, it becomes the strongest version of the universality claim.
6. If Track 2 survives grammar-generated controls, write it as a separate diagnostics paper rather than merging it into Track 1.
7. If Track 2 weakens under controls, keep it as a negative/diagnostic section in a broader research note.

Code listing: run_lie_algebraic_identities.py (1)

```
import gc
import json
import os
import warnings
from pathlib import Path

import numpy as np
import pandas as pd
import torch
import matplotlib.pyplot as plt

from transformers import AutoTokenizer, AutoModel
from sklearn.decomposition import PCA
from sklearn.metrics.pairwise import cosine_similarity

warnings.filterwarnings("ignore")


class LieAlgebraicIdentitiesAudit:
    def __init__(self):
        self.out_dir = Path(os.getenv("LIE_ALGEBRA_OUT_DIR", "results/experiments/lie_algebraic_identities_results"))
        self.csv_dir = self.out_dir / "csv"
        self.fig_dir = self.out_dir / "figures"
        self.csv_dir.mkdir(parents=True, exist_ok=True)
        self.fig_dir.mkdir(parents=True, exist_ok=True)

        self.device = "cuda" if torch.cuda.is_available() else "cpu"
        self.hf_token = None
        self.pca_dim = 64
        self.n_jacobi_null = 1000
        self.n_bootstrap = 2000

        self.models = [
            model.strip()
            for model in os.getenv(
                "LIE_ALGEBRA_MODELS",
                "bert-base-uncased,distilroberta-base,roberta-base",
            ).split(",")
            if model.strip()
        ]

        self.ops = ["N", "Q", "M", "T"]

        self.subjects = [
            "scientist", "engineer", "teacher", "doctor", "programmer",
            "researcher", "analyst", "manager", "wizard", "dragon",
            "queen", "robot", "pirate", "oracle", "knight", "alien",
        ]

        self.actions = [
            ("accepted", "accept", "the explanation"),
            ("completed", "complete", "the repair"),
            ("confirmed", "confirm", "the answer"),
            ("approved", "approve", "the treatment"),
```

Code listing: run_lie_algebraic_identities.py (2)

```
            ("fixed", "fix", "the bug"),
            ("supported", "support", "the theory"),
            ("verified", "verify", "the report"),
            ("guarded", "guard", "the treasure"),
            ("opened", "open", "the portal"),
            ("signed", "sign", "the treaty"),
        ]

    def log(self, msg):
        print(msg, flush=True)

    def base_sentence(self, subject, past, obj):
        return f"The {subject} {past} {obj}."

    def compose(self, seq, subject, past, base, obj):
        s = "".join(seq)

        mapping = {
            "": f"The {subject} {past} {obj}.",

            "N": f"The {subject} failed to {base} {obj}.",
            "Q": f"Could the {subject} {base} {obj}?",
            "M": f"The {subject} allegedly {past} {obj}.",
            "T": f"The {subject} will {base} {obj} tomorrow.",

            "NQ": f"Could the {subject} fail to {base} {obj}?",
            "QN": f"Is it false that the {subject} {past} {obj}?",

            "NM": f"The {subject} allegedly failed to {base} {obj}.",
            "MN": f"Allegedly, the {subject} failed to {base} {obj}.",

            "NT": f"The {subject} will fail to {base} {obj} tomorrow.",
            "TN": f"The {subject} failed to have {past} {obj} earlier.",

            "QM": f"Could the {subject} allegedly {base} {obj}?",
            "MQ": f"Is it alleged that the {subject} {past} {obj}?",

            "QT": f"Could the {subject} {base} {obj} tomorrow?",
            "TQ": f"Will the {subject} {base} {obj} tomorrow?",

            "MT": f"The {subject} will allegedly {base} {obj} tomorrow.",
            "TM": f"The {subject} was allegedly going to {base} {obj}.",
        }

        if s in mapping:
            return mapping[s]

        # canonical triple templates
        if s == "NQM":
            return f"Could the {subject} allegedly fail to {base} {obj}?"
        if s == "QMN":
            return f"Is it false that the {subject} allegedly {past} {obj}?"
        if s == "MNQ":
            return f"Is it alleged that the {subject} failed to {base} {obj}?"
```

Code listing: run_lie_algebraic_identities.py (3)

```

        if s == "QNM":
            return f"Could it be false that the {subject} allegedly {past} {obj}?"
        if s == "NMQ":
            return f"Could it be alleged that the {subject} failed to {base} {obj}?"
        if s == "MQN":
            return f"Allegedly, is it false that the {subject} {past} {obj}?"

        if s == "NMT":
            return f"The {subject} will allegedly fail to {base} {obj} tomorrow."
        if s == "MTN":
            return f"The {subject} was allegedly going to fail to {base} {obj}."
        if s == "TNM":
            return f"Earlier, the {subject} allegedly failed to have {past} {obj}."

        if s == "NTM":
            return f"Allegedly, the {subject} will fail to {base} {obj} tomorrow."
        if s == "TMN":
            return f"Earlier, it was alleged that the {subject} failed to have {past} {obj}."
        if s == "MNT":
            return f"It is alleged that the {subject} will fail to {base} {obj} tomorrow."

        if s == "NQT":
            return f"Could the {subject} fail to {base} {obj} tomorrow?"
        if s == "QTN":
            return f"Is it false that the {subject} will {base} {obj} tomorrow?"
        if s == "TNQ":
            return f"Could the {subject} have failed to {base} {obj} earlier?"

        if s == "QNT":
            return f"Could it be false that the {subject} will {base} {obj} tomorrow?"
        if s == "NTQ":
            return f"Could it be that the {subject} will fail to {base} {obj} tomorrow?"
        if s == "TQN":
            return f"Will it be false that the {subject} {base} {obj} tomorrow?"

        if s == "QMT":
            return f"Could the {subject} allegedly {base} {obj} tomorrow?"
        if s == "MTQ":
            return f"Could it be alleged that the {subject} will {base} {obj} tomorrow?"
        if s == "TQM":
            return f"Allegedly, will the {subject} {base} {obj} tomorrow?"

        if s == "MQT":
            return f"Will it be alleged that the {subject} {base} {obj} tomorrow?"
        if s == "QTM":
            return f"Allegedly, could the {subject} {base} {obj} tomorrow?"
        if s == "TMQ":
            return f"Could the {subject} have allegedly planned to {base} {obj}?"

        raise ValueError(f"Unsupported composition: {s}")

    def build_dataset(self, n_templates=100, seed=42):
        rng = np.random.default_rng(seed)
```

Code listing: run_lie_algebraic_identities.py (4)

```
        combos = [(s, a) for s in self.subjects for a in self.actions]
        rng.shuffle(combos)
        combos = combos[:n_templates]

        rows = []

        pairs = [("N", "Q"), ("N", "M"), ("N", "T"), ("Q", "M"), ("Q", "T"), ("M", "T")]
        triples = [("N", "Q", "M"), ("N", "Q", "T"), ("N", "M", "T"), ("Q", "M", "T")]

        for i, (subject, action) in enumerate(combos):
            past, base, obj = action
            source = self.base_sentence(subject, past, obj)

            for a, b in pairs:
                rows.append({
                    "template_id": i,
                    "kind": "antisymmetry",
                    "source": source,
                    "a": a,
                    "b": b,
                    "c": "",
                    "ab": self.compose([a, b], subject, past, base, obj),
                    "ba": self.compose([b, a], subject, past, base, obj),
                })

            for a, b, c in triples:
                rows.append({
                    "template_id": i,
                    "kind": "jacobi",
                    "source": source,
                    "a": a,
                    "b": b,
                    "c": c,
                    "abc": self.compose([a, b, c], subject, past, base, obj),
                    "bca": self.compose([b, c, a], subject, past, base, obj),
                    "cab": self.compose([c, a, b], subject, past, base, obj),
                    "acb": self.compose([a, c, b], subject, past, base, obj),
                    "cba": self.compose([c, b, a], subject, past, base, obj),
                    "bac": self.compose([b, a, c], subject, past, base, obj),
                })

        df = pd.DataFrame(rows)
        df.to_csv(self.csv_dir / "lie_algebraic_identities_dataset.csv", index=False)
        return df

    @torch.no_grad()
    def get_embeddings(self, model_name, texts, batch_size=16):
        tokenizer = AutoTokenizer.from_pretrained(model_name, token=self.hf_token)

        if tokenizer.pad_token is None:
            tokenizer.pad_token = tokenizer.eos_token or tokenizer.unk_token or "[PAD]"

        model = AutoModel.from_pretrained(model_name, token=self.hf_token).to(self.device)
        model.eval()
```

Code listing: run_lie_algebraic_identities.py (5)

```

        embs = []

        for i in range(0, len(texts), batch_size):
            batch = texts[i:i + batch_size]

            enc = tokenizer(
                batch,
                padding=True,
                truncation=True,
                max_length=128,
                return_tensors="pt",
            ).to(self.device)

            out = model(**enc)
            hidden = out.last_hidden_state
            mask = enc["attention_mask"].unsqueeze(-1)

            pooled = (hidden * mask).sum(dim=1) / mask.sum(dim=1).clamp(min=1)
            embs.append(pooled.detach().cpu().numpy())

        del model
        gc.collect()

        if self.device == "cuda":
            torch.cuda.empty_cache()

        return np.vstack(embs)

    def cos(self, x, y):
        return float(cosine_similarity(x.reshape(1, -1), y.reshape(1, -1))[0, 0])

    def alternating_null_stats(self, vectors, observed_norm, scale, rng):
        null_relative = []
        n = len(vectors)

        for _ in range(self.n_jacobi_null):
            positive = set(rng.choice(n, size=n // 2, replace=False).tolist())
            signed_sum = np.zeros_like(vectors[0])

            for i, vector in enumerate(vectors):
                signed_sum += vector if i in positive else -vector

            null_relative.append(np.linalg.norm(signed_sum) / scale)

        null_relative = np.asarray(null_relative, dtype=float)
        observed_relative = observed_norm / scale

        return {
            "null_mean_relative_jacobi_norm": float(null_relative.mean()),
            "null_std_relative_jacobi_norm": float(null_relative.std(ddof=1)),
            "jacobi_to_null_mean_ratio": float(observed_relative / (null_relative.mean() + 1e-12)),
            "null_percentile_smaller_is_better": float((null_relative <= observed_relative).mean()),
        }
```

Code listing: run_lie_algebraic_identities.py (6)

```
    def bootstrap_mean_ci(self, values, rng, alpha=0.05):
        values = np.asarray(values, dtype=float)
        if len(values) == 0:
            return np.nan, np.nan

        draws = rng.choice(values, size=(self.n_bootstrap, len(values)), replace=True).mean(axis=1)
        return (
            float(np.quantile(draws, alpha / 2)),
            float(np.quantile(draws, 1 - alpha / 2)),
        )

    def compute_for_model(self, model_name, df):
        self.log("\n" + "=" * 80)
        self.log(f"MODEL: {model_name}")
        self.log("=" * 80)

        text_cols = ["source", "ab", "ba", "abc", "bca", "cab", "acb", "cba", "bac"]
        all_texts = set()

        for col in text_cols:
            if col in df.columns:
                all_texts |= set(df[col].dropna().values)

        all_texts = sorted(all_texts)
        idx = {t: i for i, t in enumerate(all_texts)}

        raw = self.get_embeddings(model_name, all_texts)

        dim = min(self.pca_dim, raw.shape[0] - 1, raw.shape[1])
        pca = PCA(n_components=dim, random_state=42)
        vecs = pca.fit_transform(raw)

        anti_rows = []
        jacobi_rows = []

        anti_df = df[df["kind"] == "antisymmetry"]

        for row in anti_df.itertuples():
            x = vecs[idx[row.source]]
            ab = vecs[idx[row.ab]]
            ba = vecs[idx[row.ba]]

            delta_ab = ab - x
            delta_ba = ba - x

            comm_ab = delta_ab - delta_ba
            comm_ba = delta_ba - delta_ab

            antisym_error = np.linalg.norm(comm_ab + comm_ba)
            comm_norm = np.linalg.norm(comm_ab)

            anti_rows.append({
                "model": model_name,
```

Code listing: run_lie_algebraic_identities.py (7)

```
                "template_id": row.template_id,
                "pair": f"{row.a}{row.b}_vs_{row.b}{row.a}",
                "a": row.a,
                "b": row.b,
                "comm_norm": float(comm_norm),
                "antisym_error": float(antisym_error),
                "relative_antisym_error": float(antisym_error / (comm_norm + 1e-12)),
                "cos_comm_ab_neg_comm_ba": self.cos(comm_ab, -comm_ba),
            })

        jacobi_df = df[df["kind"] == "jacobi"]
        seed = sum((i + 1) * ord(ch) for i, ch in enumerate(model_name)) % (2 ** 32)
        rng = np.random.default_rng(seed)

        for row in jacobi_df.itertuples():
            x = vecs[idx[row.source]]

            abc = vecs[idx[row.abc]] - x
            bca = vecs[idx[row.bca]] - x
            cab = vecs[idx[row.cab]] - x

            acb = vecs[idx[row.acb]] - x
            cba = vecs[idx[row.cba]] - x
            bac = vecs[idx[row.bac]] - x

            # Jacobi-like alternating sum over permutations:
            # J = ABC + BCA + CAB - ACB - CBA - BAC
            jacobi = abc + bca + cab - acb - cba - bac

            positive_norm = np.linalg.norm(abc) + np.linalg.norm(bca) + np.linalg.norm(cab)
            negative_norm = np.linalg.norm(acb) + np.linalg.norm(cba) + np.linalg.norm(bac)
            scale = 0.5 * (positive_norm + negative_norm) + 1e-12

            jacobi_norm = np.linalg.norm(jacobi)
            null_stats = self.alternating_null_stats(
                [abc, bca, cab, acb, cba, bac],
                jacobi_norm,
                scale,
                rng,
            )

            jacobi_rows.append({
                "model": model_name,
                "template_id": row.template_id,
                "triple": f"{row.a}{row.b}{row.c}",
                "a": row.a,
                "b": row.b,
                "c": row.c,
                "jacobi_norm": float(jacobi_norm),
                "relative_jacobi_norm": float(jacobi_norm / scale),
                **null_stats,
            })

        anti_result = pd.DataFrame(anti_rows)
```

Code listing: run_lie_algebraic_identities.py (8)

```
        jacobi_result = pd.DataFrame(jacobi_rows)

        anti_result.to_csv(self.csv_dir / f"antisymmetry_raw_{self.safe_name(model_name)}.csv", index=False)
        jacobi_result.to_csv(self.csv_dir / f"jacobi_raw_{self.safe_name(model_name)}.csv", index=False)

        return anti_result, jacobi_result

    def safe_name(self, model_name):
        return model_name.replace("/", "_").replace("-", "_")

    def summarize(self, anti_all, jacobi_all):
        anti_summary = (
            anti_all.groupby(["model", "pair"])
            .agg(
                mean_comm_norm=("comm_norm", "mean"),
                mean_relative_antisym_error=("relative_antisym_error", "mean"),
                mean_cos_comm_ab_neg_comm_ba=("cos_comm_ab_neg_comm_ba", "mean"),
                n=("pair", "count"),
            )
            .reset_index()
        )

        jacobi_summary = (
            jacobi_all.groupby(["model", "triple"])
            .agg(
                mean_jacobi_norm=("jacobi_norm", "mean"),
                mean_relative_jacobi_norm=("relative_jacobi_norm", "mean"),
                mean_null_relative_jacobi_norm=("null_mean_relative_jacobi_norm", "mean"),
                mean_jacobi_to_null_mean_ratio=("jacobi_to_null_mean_ratio", "mean"),
                mean_null_percentile_smaller_is_better=("null_percentile_smaller_is_better", "mean"),
                n=("triple", "count"),
            )
            .reset_index()
        )

        rng = np.random.default_rng(20260611)
        ci_rows = []

        for row in jacobi_summary.itertuples():
            group = jacobi_all[
                (jacobi_all["model"] == row.model)
                & (jacobi_all["triple"] == row.triple)
            ]

            rel_low, rel_high = self.bootstrap_mean_ci(group["relative_jacobi_norm"].values, rng)
            ratio_low, ratio_high = self.bootstrap_mean_ci(group["jacobi_to_null_mean_ratio"].values, rng)

            ci_rows.append({
                "model": row.model,
                "triple": row.triple,
                "relative_jacobi_norm_ci95_low": rel_low,
                "relative_jacobi_norm_ci95_high": rel_high,
                "jacobi_to_null_mean_ratio_ci95_low": ratio_low,
                "jacobi_to_null_mean_ratio_ci95_high": ratio_high,
```

Code listing: run_lie_algebraic_identities.py (9)

```
            })

        jacobi_summary = jacobi_summary.merge(pd.DataFrame(ci_rows), on=["model", "triple"], how="left")

        anti_summary.to_csv(self.csv_dir / "antisymmetry_summary.csv", index=False)
        jacobi_summary.to_csv(self.csv_dir / "jacobi_summary.csv", index=False)

        return anti_summary, jacobi_summary

    def plot_heatmap(self, df, index_col, value_col, filename, title):
        rows = sorted(df["model"].unique())
        cols = sorted(df[index_col].unique())

        mat = np.zeros((len(rows), len(cols)))

        for i, model in enumerate(rows):
            for j, col in enumerate(cols):
                val = df[(df["model"] == model) & (df[index_col] == col)][value_col].values
                mat[i, j] = val[0] if len(val) else np.nan

        plt.figure(figsize=(10, 5))
        plt.imshow(mat, aspect="auto")
        plt.colorbar(label=value_col)

        plt.xticks(np.arange(len(cols)), cols, rotation=35, ha="right")
        plt.yticks(np.arange(len(rows)), rows)
        plt.title(title)

        for i in range(mat.shape[0]):
            for j in range(mat.shape[1]):
                plt.text(j, i, f"{mat[i, j]:.2f}", ha="center", va="center")

        path = self.fig_dir / filename
        plt.tight_layout()
        plt.savefig(path, dpi=220, bbox_inches="tight")
        plt.close()
        self.log(f"Saved figure: {path}")

    def run(self):
        self.log(f"DEVICE: {self.device}")
        self.log(f"OUT DIR: {self.out_dir}")

        df = self.build_dataset(n_templates=100, seed=42)

        anti_results = []
        jacobi_results = []

        for model in self.models:
            anti, jacobi = self.compute_for_model(model, df)
            anti_results.append(anti)
            jacobi_results.append(jacobi)

        anti_all = pd.concat(anti_results, ignore_index=True)
        jacobi_all = pd.concat(jacobi_results, ignore_index=True)
```

Code listing: run_lie_algebraic_identities.py (10)

```

        anti_all.to_csv(self.csv_dir / "antisymmetry_raw_all_models.csv", index=False)
        jacobi_all.to_csv(self.csv_dir / "jacobi_raw_all_models.csv", index=False)

        anti_summary, jacobi_summary = self.summarize(anti_all, jacobi_all)

        self.plot_heatmap(
            anti_summary,
            "pair",
            "mean_cos_comm_ab_neg_comm_ba",
            "01_antisymmetry_cosine_heatmap.png",
            "Antisymmetry Check: cos([A,B], -[B,A])",
        )

        self.plot_heatmap(
            anti_summary,
            "pair",
            "mean_relative_antisym_error",
            "02_antisymmetry_error_heatmap.png",
            "Antisymmetry Error",
        )

        self.plot_heatmap(
            jacobi_summary,
            "triple",
            "mean_relative_jacobi_norm",
            "03_jacobi_relative_norm_heatmap.png",
            "Jacobi-like Alternating Sum Relative Norm",
        )

        self.plot_heatmap(
            jacobi_summary,
            "triple",
            "mean_jacobi_to_null_mean_ratio",
            "04_jacobi_vs_permutation_null_heatmap.png",
            "Jacobi-like Norm / Permutation Null Mean",
        )

        metadata = {
            "antisymmetry": "checks whether [A,B] ≈ -[B,A]",
            "jacobi_like": "checks alternating sum ABC+BCA+CAB-ACB-CBA-BAC",
            "jacobi_null": (
                "compares the observed alternating sum with random 3-positive/3-negative "
                "sign assignments over the same six composition vectors; lower ratio and "
                "lower percentile mean stronger Jacobi-like cancellation"
            ),
            "caution": (
                "A literal nested-commutator Jacobi expansion over the same six embedded "
                "composition endpoints cancels by construction, so this script reports a "
                "non-tautological alternating third-order cancellation diagnostic instead."
            ),
            "note": "This is an empirical Lie-style diagnostic, not a formal proof of Lie algebra.",
            "models": self.models,
            "n_jacobi_null": self.n_jacobi_null,
```

Code listing: run_lie_algebraic_identities.py (11)

```
            "n_bootstrap": self.n_bootstrap,
            "jacobi_triples": ["NQM", "NQT", "NMT", "QMT"],
        }

        with open(self.out_dir / "metadata.json", "w", encoding="utf-8") as f:
            json.dump(metadata, f, indent=2)

        self.log("\nANTISYMMETRY SUMMARY")
        self.log(str(anti_summary))

        self.log("\nJACOBI SUMMARY")
        self.log(str(jacobi_summary))

        self.log("\nDONE")


if __name__ == "__main__":
    audit = LieAlgebraicIdentitiesAudit()
    audit.run()
```

Code listing: run_syntax_representation_ablation.py (1)

```
import gc
import os
from pathlib import Path

import numpy as np
import pandas as pd

from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.svm import LinearSVC

import lie_llm_syntax_holdout_experiment as syntax


OUT_DIR = Path(os.getenv("SYNTAX_ABLATION_OUT_DIR", "results/experiments/syntax_representation_ablation_results"))
OUT_DIR.mkdir(parents=True, exist_ok=True)

MODELS = [m.strip() for m in os.getenv(
    "SYNTAX_ABLATION_MODELS",
    "bert-base-uncased,roberta-base,distilroberta-base,gpt2,distilgpt2",
).split(",") if m.strip()]

N_BASE = int(os.getenv("SYNTAX_ABLATION_N_BASE", str(syntax.N_BASE)))


def run_classifiers(X, labels, train_mask, test_mask):
    classifiers = {
        "logreg": make_pipeline(
            StandardScaler(),
            LogisticRegression(max_iter=5000, class_weight="balanced", n_jobs=-1),
        ),
        "linear_svc": make_pipeline(
            StandardScaler(),
            LinearSVC(class_weight="balanced", max_iter=20000),
        ),
    }

    rows = []
    for clf_name, clf in classifiers.items():
        clf.fit(X[train_mask], labels[train_mask])
        pred = clf.predict(X[test_mask])
        rows.append({
            "classifier": clf_name,
            "accuracy": float(accuracy_score(labels[test_mask], pred)),
            "n_train": int(train_mask.sum()),
            "n_test": int(test_mask.sum()),
        })

    return rows


def main():
```

Code listing: run_syntax_representation_ablation.py (2)

```
    syntax.OUT_DIR = str(OUT_DIR)
    os.makedirs(syntax.OUT_DIR, exist_ok=True)

    base_df = syntax.make_base_rows(N_BASE)
    pair_df = syntax.build_pairs(base_df)
    prompts, pair_df = syntax.index_prompts(pair_df)

    base_df.to_csv(OUT_DIR / "base_sentences.csv", index=False)
    pair_df.to_csv(OUT_DIR / "transformation_pairs.csv", index=False)
    pd.DataFrame({"prompt_id": range(len(prompts)), "prompt": prompts}).to_csv(
        OUT_DIR / "prompts.csv",
        index=False,
    )

    labels = pair_df["class"].astype(str).to_numpy()
    syntax_family = pair_df["syntax_family"].astype(str).to_numpy()
    train_mask = np.isin(syntax_family, ["simple_svo", "with_context"])
    test_mask = np.isin(syntax_family, ["relative_clause", "temporal_clause", "reported_statement", "conditional"])
    src_idx = pair_df["source_idx"].to_numpy()
    tgt_idx = pair_df["target_idx"].to_numpy()

    all_rows = []

    for model_name in MODELS:
        print(f"\n=== MODEL: {model_name} ===")
        vectors = syntax.get_vectors(model_name, prompts)

        x_src = vectors[src_idx]
        x_tgt = vectors[tgt_idx]
        representations = {
            "x_only": x_src,
            "y_only": x_tgt,
            "delta": x_tgt - x_src,
            "concat": np.concatenate([x_src, x_tgt], axis=1),
        }

        for rep_name, X in representations.items():
            for row in run_classifiers(X, labels, train_mask, test_mask):
                row.update({
                    "model": model_name,
                    "representation": rep_name,
                    "random_baseline": 1 / len(set(labels)),
                    "n_base": N_BASE,
                })
                print(model_name, rep_name, row["classifier"], f"{row['accuracy']:.4f}")
                all_rows.append(row)

        del vectors, x_src, x_tgt, representations
        gc.collect()

    result = pd.DataFrame(all_rows)
    result.to_csv(OUT_DIR / "syntax_representation_ablation.csv", index=False)

    pivot = result.pivot_table(
```

Code listing: run_syntax_representation_ablation.py (3)

```
        index=["model", "classifier"],
        columns="representation",
        values="accuracy",
    ).reset_index()
    pivot.to_csv(OUT_DIR / "syntax_representation_ablation_pivot.csv", index=False)

    print("\nSUMMARY")
    print(pivot)
    print("Saved to", OUT_DIR)


if __name__ == "__main__":
    main()
```

Code listing: run_layerwise_pooling_ablation.py (1)

```
import gc
import os
import time
from pathlib import Path

import numpy as np
import pandas as pd
import torch
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.svm import LinearSVC
from transformers import AutoModel, AutoTokenizer

import lie_llm_syntax_holdout_experiment as syntax


OUT_DIR = Path(os.getenv("LAYERWISE_OUT_DIR", "results/experiments/layerwise_pooling_ablation_results"))
OUT_DIR.mkdir(parents=True, exist_ok=True)

MODEL_NAME = os.getenv("LAYERWISE_MODEL", "bert-base-uncased")
N_BASE = int(os.getenv("LAYERWISE_N_BASE", "300"))
BATCH_SIZE = int(os.getenv("LAYERWISE_BATCH_SIZE", "16"))


def pool(hidden, mask, mode):
    if mode == "mean":
        m = mask.unsqueeze(-1).float()
        return (hidden * m).sum(dim=1) / m.sum(dim=1).clamp_min(1e-9)
    if mode == "cls":
        return hidden[:, 0, :]
    if mode == "last_token":
        idx = mask.sum(dim=1).clamp_min(1) - 1
        return hidden[torch.arange(hidden.shape[0]), idx]
    raise ValueError(mode)


def extract_layer_vectors(prompts):
    tokenizer = AutoTokenizer.from_pretrained(MODEL_NAME)
    if tokenizer.pad_token is None:
        tokenizer.pad_token = tokenizer.eos_token or tokenizer.unk_token or "[PAD]"

    model = AutoModel.from_pretrained(MODEL_NAME, output_hidden_states=True)
    model.eval()

    rows_by_key = {}
    started = time.time()

    for start in range(0, len(prompts), BATCH_SIZE):
        batch = prompts[start:start + BATCH_SIZE]
        enc = tokenizer(batch, padding=True, truncation=True, max_length=160, return_tensors="pt")

        with torch.no_grad():
```

Code listing: run_layerwise_pooling_ablation.py (2)

```
            out = model(**enc)

        for layer_idx, hidden in enumerate(out.hidden_states):
            for pooling in ["mean", "cls", "last_token"]:
                key = (layer_idx, pooling)
                rows_by_key.setdefault(key, []).append(pool(hidden, enc["attention_mask"], pooling).cpu().numpy())

        done = start + len(batch)
        print(f"{MODEL_NAME}: {done}/{len(prompts)}, {time.time() - started:.1f}s")

    result = {key: np.vstack(parts) for key, parts in rows_by_key.items()}

    del model
    gc.collect()
    return result


def fit_score(X, labels, train_mask, test_mask):
    classifiers = {
        "logreg": make_pipeline(StandardScaler(), LogisticRegression(max_iter=5000, class_weight="balanced", n_jobs=-1)),
        "linear_svc": make_pipeline(StandardScaler(), LinearSVC(class_weight="balanced", max_iter=20000)),
    }

    rows = []
    for clf_name, clf in classifiers.items():
        clf.fit(X[train_mask], labels[train_mask])
        pred = clf.predict(X[test_mask])
        rows.append({
            "classifier": clf_name,
            "accuracy": float(accuracy_score(labels[test_mask], pred)),
        })
    return rows


def main():
    base_df = syntax.make_base_rows(N_BASE)
    pair_df = syntax.build_pairs(base_df)
    prompts, pair_df = syntax.index_prompts(pair_df)

    pair_df.to_csv(OUT_DIR / "transformation_pairs.csv", index=False)
    pd.DataFrame({"prompt_id": range(len(prompts)), "prompt": prompts}).to_csv(OUT_DIR / "prompts.csv", index=False)

    labels = pair_df["class"].astype(str).to_numpy()
    syntax_family = pair_df["syntax_family"].astype(str).to_numpy()
    train_mask = np.isin(syntax_family, ["simple_svo", "with_context"])
    test_mask = np.isin(syntax_family, ["relative_clause", "temporal_clause", "reported_statement", "conditional"])
    src_idx = pair_df["source_idx"].to_numpy()
    tgt_idx = pair_df["target_idx"].to_numpy()

    vectors_by_key = extract_layer_vectors(prompts)

    all_rows = []
    for (layer_idx, pooling), vectors in vectors_by_key.items():
        x_src = vectors[src_idx]
```

Code listing: run_layerwise_pooling_ablation.py (3)

```
        x_tgt = vectors[tgt_idx]

        for rep_name, X in {
            "delta": x_tgt - x_src,
            "y_only": x_tgt,
        }.items():
            for row in fit_score(X, labels, train_mask, test_mask):
                row.update({
                    "model": MODEL_NAME,
                    "layer": layer_idx,
                    "pooling": pooling,
                    "representation": rep_name,
                    "n_base": N_BASE,
                    "n_train": int(train_mask.sum()),
                    "n_test": int(test_mask.sum()),
                })
                print(row)
                all_rows.append(row)

    result = pd.DataFrame(all_rows)
    result.to_csv(OUT_DIR / "layerwise_pooling_ablation.csv", index=False)

    best = (
        result[result["classifier"] == "linear_svc"]
        .sort_values("accuracy", ascending=False)
        .head(20)
    )
    best.to_csv(OUT_DIR / "layerwise_pooling_ablation_top20.csv", index=False)
    print(best)
    print("Saved to", OUT_DIR)


if __name__ == "__main__":
    main()
```

Code listing: run_track1_spotcheck.py (1)

```
import gc
import os
from pathlib import Path

import numpy as np
import pandas as pd
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.svm import LinearSVC

import lie_llm_full_semantic_holdout_experiment as full


OUT_DIR = Path(os.getenv("SPOTCHECK_OUT_DIR", "results/experiments/track1_spotcheck_results"))
OUT_DIR.mkdir(parents=True, exist_ok=True)

MODELS = [m.strip() for m in os.getenv("SPOTCHECK_MODELS", "bert-large-uncased").split(",") if m.strip()]
N_BASE = int(os.getenv("SPOTCHECK_N_BASE", "100"))


def run_classifiers(X, labels, split):
    train_mask = split == "train"
    test_mask = split == "test"
    classifiers = {
        "logreg": make_pipeline(StandardScaler(), LogisticRegression(max_iter=5000, class_weight="balanced", n_jobs=-1)),
        "linear_svc": make_pipeline(StandardScaler(), LinearSVC(class_weight="balanced", max_iter=20000)),
    }
    rows = []
    for clf_name, clf in classifiers.items():
        clf.fit(X[train_mask], labels[train_mask])
        pred = clf.predict(X[test_mask])
        rows.append({
            "classifier": clf_name,
            "accuracy": float(accuracy_score(labels[test_mask], pred)),
            "n_train": int(train_mask.sum()),
            "n_test": int(test_mask.sum()),
        })
    return rows


def main():
    full.OUT_DIR = str(OUT_DIR)
    full.N_BASE = N_BASE
    os.makedirs(full.OUT_DIR, exist_ok=True)

    pair_df = full.build_dataset()
    prompts, pair_df = full.index_prompts(pair_df)
    pair_df.to_csv(OUT_DIR / "transformation_pairs.csv", index=False)
    pd.DataFrame({"prompt_id": range(len(prompts)), "prompt": prompts}).to_csv(OUT_DIR / "prompts.csv", index=False)

    labels = pair_df["class"].astype(str).to_numpy()
    split = pair_df["split"].astype(str).to_numpy()
```

Code listing: run_track1_spotcheck.py (2)

```
    src_idx = pair_df["source_idx"].to_numpy()
    tgt_idx = pair_df["target_idx"].to_numpy()

    rows = []
    for model_name in MODELS:
        print(f"\n=== MODEL: {model_name} ===")
        vectors = full.get_vectors(model_name, prompts)
        x_src = vectors[src_idx]
        x_tgt = vectors[tgt_idx]
        reps = {
            "x_only": x_src,
            "y_only": x_tgt,
            "delta": x_tgt - x_src,
            "concat": np.concatenate([x_src, x_tgt], axis=1),
        }
        for rep_name, X in reps.items():
            for row in run_classifiers(X, labels, split):
                row.update({
                    "model": model_name,
                    "representation": rep_name,
                    "random_baseline": 1 / len(set(labels)),
                    "n_base": N_BASE,
                })
                print(model_name, rep_name, row["classifier"], f"{row['accuracy']:.4f}")
                rows.append(row)
        del vectors, reps
        gc.collect()

    result = pd.DataFrame(rows)
    result.to_csv(OUT_DIR / "spotcheck_representation_ablation.csv", index=False)
    pivot = result.pivot_table(index=["model", "classifier"], columns="representation", values="accuracy").reset_index()
    pivot.to_csv(OUT_DIR / "spotcheck_representation_ablation_pivot.csv", index=False)
    print(pivot)


if __name__ == "__main__":
    main()
```

Code listing: run_full_semantic_pooling_ablation.py (1)

```
import gc
import os
import random
from pathlib import Path

import numpy as np
import pandas as pd
import torch
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.svm import LinearSVC
from transformers import AutoModel, AutoTokenizer

import lie_llm_full_semantic_holdout_experiment as full


OUT_DIR = Path(os.getenv("POOLING_OUT_DIR", "results/experiments/full_semantic_pooling_ablation_results"))
MODELS = [m.strip() for m in os.getenv("POOLING_MODELS", "bert-base-uncased,roberta-base,gpt2").split(",") if m.strip()]
N_BASE = int(os.getenv("POOLING_N_BASE", "150"))
BATCH_SIZE = int(os.getenv("POOLING_BATCH_SIZE", "32"))
SEED = int(os.getenv("POOLING_SEED", "42"))


def reset_full_dataset_seed():
    full.N_BASE = N_BASE
    full.random.seed(SEED)
    full.np.random.seed(SEED)
    random.seed(SEED)
    np.random.seed(SEED)
    torch.manual_seed(SEED)


def safe_name(model_name):
    return model_name.replace("/", "__")


def last_token_pool(hidden, attention_mask):
    lengths = attention_mask.sum(dim=1).clamp(min=1) - 1
    batch_idx = torch.arange(hidden.shape[0], device=hidden.device)
    return hidden[batch_idx, lengths]


def get_pooled_vectors(model_name, prompts):
    tokenizer = AutoTokenizer.from_pretrained(model_name)
    if tokenizer.pad_token is None:
        tokenizer.pad_token = tokenizer.eos_token or tokenizer.unk_token or "[PAD]"

    model = AutoModel.from_pretrained(model_name, output_hidden_states=True)
    model.eval()

    pooled = {"mean": [], "cls": [], "last_token": []}

```

Code listing: run_full_semantic_pooling_ablation.py (2)

```
    for start in range(0, len(prompts), BATCH_SIZE):
        batch = prompts[start:start + BATCH_SIZE]
        inputs = tokenizer(
            batch,
            padding=True,
            truncation=True,
            max_length=180,
            return_tensors="pt",
        )

        with torch.no_grad():
            out = model(**inputs)

        hidden = out.hidden_states[-1]
        mask = inputs["attention_mask"]

        pooled["mean"].append(full.mean_pool(hidden, mask).cpu().numpy())
        pooled["cls"].append(hidden[:, 0, :].cpu().numpy())
        pooled["last_token"].append(last_token_pool(hidden, mask).cpu().numpy())

        done = start + len(batch)
        print(f"  {model_name}: {done}/{len(prompts)}", flush=True)

    del model
    gc.collect()

    return {k: np.vstack(v) for k, v in pooled.items()}


def build_features(vectors, pair_df, representation):
    src = vectors[pair_df["source_idx"].to_numpy()]
    tgt = vectors[pair_df["target_idx"].to_numpy()]

    if representation == "x_only":
        return src
    if representation == "y_only":
        return tgt
    if representation == "delta":
        return tgt - src
    if representation == "concat":
        return np.hstack([src, tgt])
    raise ValueError(representation)


def run_one(model_name, pooling, representation, vectors, pair_df):
    X = build_features(vectors, pair_df, representation)
    y = pair_df["class"].astype(str).to_numpy()
    split = pair_df["split"].astype(str).to_numpy()

    train = split == "train"
    test = split == "test"

    classifiers = {
        "logreg": make_pipeline(
```

Code listing: run_full_semantic_pooling_ablation.py (3)

```
            StandardScaler(),
            LogisticRegression(max_iter=5000, class_weight="balanced"),
        ),
        "linear_svc": make_pipeline(
            StandardScaler(),
            LinearSVC(class_weight="balanced", max_iter=20000),
        ),
    }

    rows = []
    for clf_name, clf in classifiers.items():
        clf.fit(X[train], y[train])
        pred = clf.predict(X[test])
        rows.append({
            "model": model_name,
            "pooling": pooling,
            "representation": representation,
            "classifier": clf_name,
            "accuracy": float(accuracy_score(y[test], pred)),
            "n_base": N_BASE,
            "n_train": int(train.sum()),
            "n_test": int(test.sum()),
        })
    return rows


def main():
    OUT_DIR.mkdir(parents=True, exist_ok=True)
    reset_full_dataset_seed()

    pair_df = full.build_dataset()
    prompts, pair_df = full.index_prompts(pair_df)
    pair_df.to_csv(OUT_DIR / "transformation_pairs.csv", index=False)
    pd.DataFrame({"prompt_id": range(len(prompts)), "prompt": prompts}).to_csv(OUT_DIR / "prompts.csv", index=False)

    rows = []
    for model_name in MODELS:
        print(f"\n=== MODEL: {model_name} ===", flush=True)
        pooled_vectors = get_pooled_vectors(model_name, prompts)

        for pooling, vectors in pooled_vectors.items():
            for representation in ["x_only", "y_only", "concat", "delta"]:
                rows.extend(run_one(model_name, pooling, representation, vectors, pair_df))

    result = pd.DataFrame(rows)
    result.to_csv(OUT_DIR / "full_semantic_pooling_ablation.csv", index=False)
    pivot = result.pivot_table(
        index=["model", "pooling", "classifier"],
        columns="representation",
        values="accuracy",
    ).reset_index()
    pivot.to_csv(OUT_DIR / "full_semantic_pooling_ablation_pivot.csv", index=False)
    print(pivot)

```

Code listing: run_full_semantic_pooling_ablation.py (4)

```

if __name__ == "__main__":
    main()
```

Code listing: analyze_confusion_negation.py (1)

```
from pathlib import Path

import pandas as pd


ROOT = Path(__file__).resolve().parents[1]
EXP = ROOT / "results" / "experiments"
OUT = ROOT / "results"
OUT.mkdir(exist_ok=True)


def parse_name(path):
    stem = path.stem.replace("confusion_", "")
    if stem.endswith("_linear_svc"):
        return stem[:-11].replace("__", "/"), "linear_svc"
    if stem.endswith("_logreg"):
        return stem[:-7].replace("__", "/"), "logreg"
    raise ValueError(path.name)


def load_confusion(path):
    model, classifier = parse_name(path)
    cm = pd.read_csv(path, index_col=0)
    labels = list(cm.index)

    rows = []
    errors = []

    for label in labels:
        total = float(cm.loc[label].sum())
        correct = float(cm.loc[label, label])
        recall = correct / total if total else 0.0
        rows.append({
            "source": "full_semantic",
            "model": model,
            "classifier": classifier,
            "class": label,
            "support": int(total),
            "correct": int(correct),
            "recall": recall,
            "error_rate": 1.0 - recall,
            "is_negation": label == "negation",
        })

        for pred in labels:
            if pred == label:
                continue
            count = int(cm.loc[label, pred])
            if count:
                errors.append({
                    "source": "full_semantic",
                    "model": model,
                    "classifier": classifier,
                    "true_class": label,
```

Code listing: analyze_confusion_negation.py (2)

```
                    "predicted_class": pred,
                    "count": count,
                    "true_support": int(total),
                    "share_of_true": count / total if total else 0.0,
                    "involves_negation": label == "negation" or pred == "negation",
                })

    return rows, errors


def main():
    paths = sorted((EXP / "lie_llm_full_semantic_holdout_results").glob("confusion_*.csv"))
    all_rows = []
    all_errors = []

    for path in paths:
        rows, errors = load_confusion(path)
        all_rows.extend(rows)
        all_errors.extend(errors)

    class_df = pd.DataFrame(all_rows)
    error_df = pd.DataFrame(all_errors)

    class_df.to_csv(OUT / "confusion_class_recall.csv", index=False)
    error_df.sort_values(["count", "share_of_true"], ascending=False).to_csv(
        OUT / "confusion_top_errors.csv",
        index=False,
    )

    neg_summary = (
        class_df
        .assign(group=lambda d: d["is_negation"].map({True: "negation", False: "non_negation"}))
        .groupby(["source", "model", "classifier", "group"])
        .agg(
            mean_recall=("recall", "mean"),
            mean_error_rate=("error_rate", "mean"),
            n_classes=("class", "count"),
        )
        .reset_index()
    )
    neg_summary.to_csv(OUT / "confusion_negation_summary.csv", index=False)

    rank_df = class_df.copy()
    rank_df["recall_rank_low_is_hard"] = rank_df.groupby(["model", "classifier"])["recall"].rank(method="min")
    rank_df.sort_values(["model", "classifier", "recall"]).to_csv(
        OUT / "confusion_class_recall_ranked.csv",
        index=False,
    )

    print("Saved:")
    for name in [
        "confusion_class_recall.csv",
        "confusion_top_errors.csv",
        "confusion_negation_summary.csv",
```

Code listing: analyze_confusion_negation.py (3)

```
        "confusion_class_recall_ranked.csv",
    ]:
        print(OUT / name)


if __name__ == "__main__":
    main()
```

Code listing: build_signed_permutation_multiple_testing.py (1)

```
from pathlib import Path

import numpy as np
import pandas as pd


ROOT = Path(__file__).resolve().parents[1]
EXP = ROOT / "results" / "experiments"
OUT = EXP / "lie_algebraic_identities_results" / "csv"


def benjamini_hochberg(p_values):
    p = np.asarray(p_values, dtype=float)
    n = len(p)
    order = np.argsort(p)
    ranked = p[order]
    adj = np.empty(n, dtype=float)
    prev = 1.0
    for i in range(n - 1, -1, -1):
        rank = i + 1
        val = min(prev, ranked[i] * n / rank)
        prev = val
        adj[order[i]] = val
    return np.clip(adj, 0, 1)


def bootstrap_p_less_than_one(values, n_boot=20000, seed=42):
    rng = np.random.default_rng(seed)
    values = np.asarray(values, dtype=float)
    idx = rng.integers(0, len(values), size=(n_boot, len(values)))
    means = values[idx].mean(axis=1)
    p = (np.sum(means >= 1.0) + 1) / (n_boot + 1)
    return float(p)


def main():
    files = [
        EXP / "lie_algebraic_identities_results" / "csv" / "jacobi_raw_all_models.csv",
        EXP / "lie_algebraic_identities_decoder_results" / "csv" / "jacobi_raw_all_models.csv",
    ]
    raw = pd.concat([pd.read_csv(path) for path in files if path.exists()], ignore_index=True)

    rows = []
    for (model, triple), group in raw.groupby(["model", "triple"]):
        ratios = group["jacobi_to_null_mean_ratio"].to_numpy()
        mean_ratio = float(ratios.mean())
        p_less = bootstrap_p_less_than_one(ratios)
        rows.append({
            "model": model,
            "triple": triple,
            "n": int(len(group)),
            "mean_ratio_to_null": mean_ratio,
            "bootstrap_p_ratio_less_than_1": p_less,
            "direction": "below_null" if mean_ratio < 1 else "above_null",
```

Code listing: build_signed_permutation_multiple_testing.py (2)

```
        })

    result = pd.DataFrame(rows).sort_values(["bootstrap_p_ratio_less_than_1", "model", "triple"])
    result["bonferroni_p"] = np.minimum(result["bootstrap_p_ratio_less_than_1"] * len(result), 1.0)
    result["bh_fdr_p"] = benjamini_hochberg(result["bootstrap_p_ratio_less_than_1"].to_numpy())
    result["passes_bonferroni_0_05"] = result["bonferroni_p"] < 0.05
    result["passes_bh_fdr_0_05"] = result["bh_fdr_p"] < 0.05

    out_path = OUT / "signed_permutation_multiple_testing.csv"
    result.to_csv(out_path, index=False)
    print(out_path)
    print(result)


if __name__ == "__main__":
    main()
```